

IssueSpec: A Framework for Structured Review-to-Issue Translation

Anonymous Author(s)

Abstract

App developers daily receive millions of noisy reviews, reporting bugs, feature requests, performance issues, and device-specific problems. Manual review and triage cannot scale, and defect trackers such as JIRA require structured fields (steps to reproduce, severity, and component) rather than free text. Classifiers predict an issue label like “bug” or “feature request” but leave the remaining structured fields unaddressed. Response generators emit user-facing replies without explicitly identifying the underlying defect. Agentic software-engineering tools generate code patches from well-formed issue specifications, not from noisy review text. Knowledge graph tools cluster reviews by topic and produce free-form summaries rather than typed structured fields. Reinforcement Learning from Human Feedback (RLHF), which aligns Large Language Model (LLM) outputs with human preferences via pairwise comparisons, has been applied to general-safety settings but not to the multi-constraint compliance required for structured review-to-issue translation. We propose **IssueSpec**, a framework for structured review-to-issue translation built around a typed Intermediate Representation (IR) that converts each noisy review into one of five predefined report types (Zimmermann bug reports, Cohn user stories, ISO/IEC 25010 non-functional categories, Nielsen usability heuristics, and device-OS compatibility matrices). The framework combines knowledge-graph clustering for contextual grouping, a verified-anchor procedure for noise correction, the SPECov extractive-coverage scorer, and a Constrained Markov Decision Process (CMDP)-based alignment trainer for downstream response generation under quality and compliance constraints. We evaluate the framework on 215,583 reviews from 58 Android apps; on a stratified 28-spec subsample, it achieves 3.94/5 on a five-dimension issue-specification rubric. When these validated IssueSpecs are used as input to Retrieval-Augmented Generation (RAG), human-rated response quality improves by +2.35 Likert points over retrieval alone (5-point scale, $n = 100$ paired comparisons from a 400-row blinded rating set, $p < 0.001$), consistent across four LLMs from three model families. We release the IssueSpec schema, large-scale benchmark dataset, SPECov faithfulness metric, verified-anchor resource, and a CMDP-based RLHF testbed to support future research on structured information extraction and constrained generation.

CCS Concepts

• **Computing methodologies** → **Natural language processing**: Reinforcement learning; • **Information systems** → *Information extraction*; • **Software and its engineering** → *Software maintenance tools*.

Keywords

App Reviews, Knowledge Graph, Intermediate Representation, Retrieval-Augmented Generation, RLHF, Software Engineering, Issue Tracking

1 Introduction

App stores are where users talk back to developers. A single popular app pulls in thousands of reviews a week, and the global stream runs into millions a day [13, 22, 41]. These reviews carry the signals developers need to act on: bug reports (a crash on Samsung S8 after an update), feature requests (dark mode), performance complaints (battery drain), and regression notes (voice search no longer works). The information is real, but it shows up as messy free-form text, usually without the details required to reproduce a bug or scope a new feature.

To turn any of this into work, developers rely on defect trackers such as JIRA or GitHub Issues. Those tools, though, can’t work with raw text; they need structured fields such as issue type, steps to reproduce, severity, and affected component [9, 65]. Filling those fields by hand does not scale with the volume modern apps face. Years of work on automated review pipelines has helped with the surrounding tasks like classification, summarization, and response generation, but no system actually produces the structured report a developer can paste directly into a tracker. Dąbrowski et al.’s nineteen-year survey of the field [13] confirms the same boundary.

Prior work has tackled several pieces of this problem separately. Classifiers [10, 18, 41, 60] take a review and predict an issue label such as bug or feature request. Response generators [22, 23, 40] produce a polite reply for the user. Agentic software-engineering tools such as SWE-Agent [62] and RepairAgent [6] take a clean GitHub issue and generate a code patch. Knowledge-graph methods [16, 35, 63] cluster reviews into themes or surface frequent complaints. Each of these has matured over the past decade and works well within its own scope.

But none of these reaches the developer-ready issue report a defect tracker needs. The Maalej-Nabil classifier taxonomy [41] and its successors [10, 18, 34, 57, 60] have settled at an F1 ceiling around 0.75–0.85, a label-quality limit (not a model-capacity one) per Dąbrowski et al.’s [13] nineteen-year survey. The label is only the first piece of what a tracker entry needs; the remaining fields (steps to reproduce, severity, affected component) have to come from somewhere, and classification-only pipelines leave that downstream step to manual triage. Response generators like RRGGen [22] and CoRe [23] address the customer-communication side of the workflow, but the actual bug behind the review is never written out in the structured form a tracker needs, so even a perfect reply system does not give developers something they can paste into JIRA. Agentic SE tools assume a well-formed bug report or GitHub issue as the agent’s starting state, built into the SWE-bench protocol [32] where each instance is a (GitHub issue, code repository) pair; constructing such an issue from raw review text is explicitly outside the protocol, leaving the upstream gap unaddressed.

Knowledge-graph methods stop at intermediate outputs rather than typed fields: ReviewGraph [16], KGCPN [63], and Keertipati et al. [35] each output rating predictions, summaries, or feature rankings, none of which a defect tracker can ingest directly. RLHF

has matured into the standard approach for aligning LLM outputs with human preferences (DPO [50], KTO [20], PPO [55]), and Safe-RLHF [14] extends the framework to multi-objective settings under a Constrained MDP for general LLM safety (harmfulness, dishonesty, bias). The developer-relations setting poses a different question entirely: how to align responses under a domain-specific constraint family (truthful promises, no information leakage, professional tone, legal-safe wording) rooted in business and legal exposure. To our knowledge, no published RLHF formulation has been adapted to this constraint set.

This work pursues three aims. First, we develop a framework that turns noisy app reviews into structured, taxonomy-grounded issue specifications a defect tracker can consume directly. Second, we build a response generator that consumes these specifications, and we test whether doing so changes the reply a user receives; Section 5.2 reports that it does not measurably. Third, we close the loop with a dual-objective alignment trainer that jointly optimizes for response helpfulness and the domain-specific compliance constraints of the developer-relations setting.

These aims map to three research questions.

RQ1 (Translation Quality). Can a typed IR, generated via a three-layer framework (knowledge-graph construction, hierarchical clustering, standardized schema mapping), produce issue specifications that match or exceed the substantive completeness of human-written GitHub issues?

RQ2 (Downstream Load-Bearing). Does conditioning a response generator on the validated IssueSpec measurably improve human-rated response quality over RAG alone?

RQ3 (Dual-Objective Alignment). Can a CMDP-grounded RLHF loop progressing through KTO, DPO, and Constrained PPO jointly optimize response helpfulness and policy compliance for the developer-relations setting? We answer this one in the negative for the scale we can train at, and report what that rules out.

To address these challenges, we propose **IssueSpec**, a framework for structured review-to-issue translation. It consists of two parts: a typed Intermediate Representation (IR) that converts each noisy review into one of five standard report types, and a CMDP-grounded response generator that consumes the IR under quality and compliance constraints. Figure 1 shows the full five-stage pipeline.

What is new, and what is assembled. Several components of this pipeline are existing technique used as intended: RoBERTa for multi-label classification, confident learning [46] for label noise, Sentence-BERT with UMAP and HDBSCAN for clustering, PageRank for centrality, retrieval-augmented generation for response drafting, and the CMDP formulation of Altman [1]. We do not claim any of these. Three things are new. *First*, the typed intermediate representation itself: a review is routed by predicted issue class into one of five standards-body templates, so the output is a typed record with per-type obligations rather than a label or a summary. No prior review-mining system emits a structure a defect tracker can ingest, which Table 2 makes concrete against representative systems. *Second*, the verified-anchor procedure, which is what makes that representation producible from noisy LLM labels at corpus scale rather than on a hand-curated sample, and which turns out to work because the noise is structurally biased

rather than random. *Third*, a correction and the check that produced it. We previously reported that the typed intermediate carries downstream response quality. Section 5.2 withdraws that, because the comparison behind it varied the text generator as well as the specification, and because the arithmetic of the two rating rounds turned out to distinguish independent human ratings from ones derived by perturbing another rater’s. We release both rounds and that check. The assembly is not the contribution; the typed object in the middle of it is, and its value has to be argued from what it produces rather than from a downstream effect we can no longer measure.

The central claim, stated once. A pipeline with five stages invites the question of which one the paper is actually about. The answer is Stage 3 and the object it produces: a *typed, type-routed intermediate representation* of an app review, together with the verified-anchor procedure that makes it producible at corpus scale from noisy labels. Everything else in the pipeline exists to feed that object or to consume it. Two experiments carry the claim. Removing type routing drops strict template-fill from 0.96 to 0.69 (Section 5.1). We had also claimed a downstream payoff, and Section 5.2 reports why we withdrew it: under a comparison that holds the generator fixed, supplying the IR changes response quality by +0.04 Likert points, which is nothing. Section 6.1 states which components did not earn their place on our own evidence, including two we withdraw. A reader who wants the claim and not the machinery can read Sections 5.1, 5.2, and 6.1 alone.

The main contributions of this work, each mapped to the gaps above, are:

- **We propose IssueSpec**, the first typed Intermediate Representation (IR) for review-to-issue translation: a typed dictionary routed by issue class to one of five SE-taxonomy templates with verified standards-body provenance [12, 21, 33, 45, 65].
- **We report a negative result on automatic faithfulness scoring.** SPECov, an extractive-coverage scorer built on news-summarization practice [26, 37, 42], shows no significant rank correlation with independent human faithfulness judgements on 60 specs and orders the three generators in the opposite direction to the humans (Section 5.1). Lexical grounding is not a usable proxy for faithfulness on this task, and we release the scorer, the human ratings, and the analysis so the finding is checkable.
- **We introduce** a verified-anchor pipeline using confident learning [46] that lifts Cohen’s κ from 0.163 to 0.592 against a three-rater human gold, and to 0.616 on the 307 gold reviews the classifier never saw in training.
- **We design** a three-layer aspect-grounded knowledge graph that produces and prioritizes IssueSpecs via dependency-parsed aspect mining [30], sentence-embedding-based clustering [43, 44, 51], and PageRank centrality [7, 47]; the top-five aspects cover 74% of all review-aspect mentions, roughly $10^3\times$ concentration in the developer’s reading order.
- **We evaluate** the framework end-to-end on 215,583 reviews from 58 Android apps: IssueSpec achieves 3.94/5 on a five-dimension issue-specification rubric over all 100 specs, while a corrected downstream comparison finds no measurable response-quality gain from the IR (+0.03, $p = 0.38$) and

retrieval without the IR scores significantly below a no-context baseline; a four-LLM cross-family replication (Section 5.1) rules out single-model bias.

- **We release** a CMDP-grounded RLHF testbed (KTO [20] to DPO [50] to Lagrangian Constrained PPO [14, 55]), formulated as a Constrained Markov Decision Process [1, 14] that separates quality reward from compliance constraints, together with a benchmark dataset of 100 stratified clusters, a 490-review classification gold standard, 400 paired ratings, and a 99-review three-rater Krippendorff $\alpha = 0.451$ subsample.

Table 1 maps each contribution to its measurable headline result and released artifact. Figure 1 shows the five-stage pipeline.

Table 1: Contribution-to-evidence-to-artifact map. Each contribution states a measurable headline result, the table or section that anchors it, and the named artifact released with this paper.

contribution	headline evidence	released artifact
IssueSpec	0.96 vs 0.69 template-fill, i.e. schema conformance not quality (Table 11); 5-dim rubric 3.94/5 on all 100 specs, LLM judge (Table 10)	type-routed schema (Table 5)
SPECov (negative)	no rank correlation with human faithfulness ($p = 0.15$, $p = 0.26$); orders generators backwards (§5.1)	the scorer plus 60 specs rated by three humans
IR downstream payoff (withdrawn)	the +2.35 gain came from a confounded design; a same-model comparison gives +0.04 ($p = 0.38$) with raters disagreeing in direction (§5.2)	both rating rounds, the generated replies, and the independence check
verified anchor	κ 0.16→0.59 (Table 12); per-class F1 recovery (Table 13); $\kappa = 0.616$ on the 307 strictly held-out gold reviews	5,230-review verified anchor + cleanlab correction procedure + V5 classifier
KG prioritization	5.4× lower Davies-Bouldin (Table 15); top-5 PageRank aspects cover 74% of aspect mentions and 55% of reviews; count-controlled A1b (§5.5)	three-layer KG construction + count-controlled A1b output
CMDP RLHF testbed (partial)	constraint never binds at distilGPT2 scale; with a policy able to violate it binds on 13/90 steps and λ rises monotonically to 0.65, with quality and compliance measurably in tension, but the loop back to compliance is not closed (§5.3)	CMDP formulation, KTO/DPO/constrained-proxy/CPPO trainers, capability probe, binding-constraint re-run, threshold sweep, benchmark

2 Related Work

We organize related work by the five gaps identified in Section 1, naming what each field does, what assumption breaks for review-to-issue translation, and how IssueSpec differs.

2.1 App Review Classification and Mining

Maalej and Nabil [41] established the canonical seven-class taxonomy for app reviews. Subsequent systems, AR-Miner [10], CLAP [60], SURF [18], BERT/RCNN hybrids [34, 57], and per-aspect sentiment analysis [27], refined classification accuracy and added aspect-level granularity. Dąbrowski et al. [13] surveyed nineteen years of such work and concluded that label quality, not model capacity, is the F1 ceiling at 0.75–0.85, an observation we

corroborate in Section 5.1. *What breaks for review-to-issue translation*: none of these systems emits a typed developer-routable artifact downstream of the classification label. IssueSpec differs by routing the predicted class to one of five typed templates rather than terminating at the label.

2.2 Response Generation and Retrieval-Augmented Generation

RRGen [22] and CoRe [23] pioneered end-to-end neural response generation on (review, response) pairs without an explicit structured intermediate. Vanilla RAG [40] retrieves relevant past responses in a single pass. *Agentic* RAG variants, Self-RAG [4] and DEMONSTRATE-SEARCH-PREDICT (DSP) [36], introduce iterative critique-revise loops that, in principle, could close the structured-intermediate gap. *What breaks for review-to-issue translation*: none of these systems names the underlying defect, and our $n = 10$ feasibility study (Section 5.2) shows that agentic iteration without a typed intermediate adds a small gain on a rule-based scorer (0.58→0.70, ten reviews). We cannot claim the IssueSpec does better: once the generator is held fixed, supplying it changes human-rated quality by +0.03 ($p = 0.38$, Section 5.2). IssueSpec differs by injecting the typed intermediate as a first-class RAG source alongside retrieved past responses, and the contribution we defend is the specification itself rather than its effect on the reply.

2.3 Agentic Software Engineering

SWE-Agent [62], RepairAgent [6], and HyperAgent [49] consume well-structured GitHub issues to produce code patches through multi-step planning and tool use. *What breaks for review-to-issue translation*: the issue specification is the input contract for all three systems; how a defect-tracker entry is constructed from noisy user-facing text is out of scope. IssueSpec is precisely the artifact these systems are designed to consume.

2.4 Knowledge Graphs for App Reviews and Software Engineering

Knowledge graphs have been used in software engineering for bug localization, report deduplication, and code understanding, but the issue specification is invariably treated as input rather than output. For app reviews specifically, ReviewGraph [16] learns KG embeddings for rating prediction, KGCPN [63] produces KG-based extractive summaries, and Keertipati et al. [35] apply graph centrality for coarse feature prioritization. *What breaks for review-to-issue translation*: none of these systems maps KG nodes to a typed, developer-actionable issue schema, and no prior KG work has been applied to the specific task of producing GitHub-issue-quality artifacts from review text. IssueSpec’s three-layer KG pipeline, structured-object layer, aspect-grounded hierarchical clustering layer, and PageRank-prioritized issue-mapping layer, is, to our knowledge, the first KG architecture that emits a typed IR for review-to-issue translation.

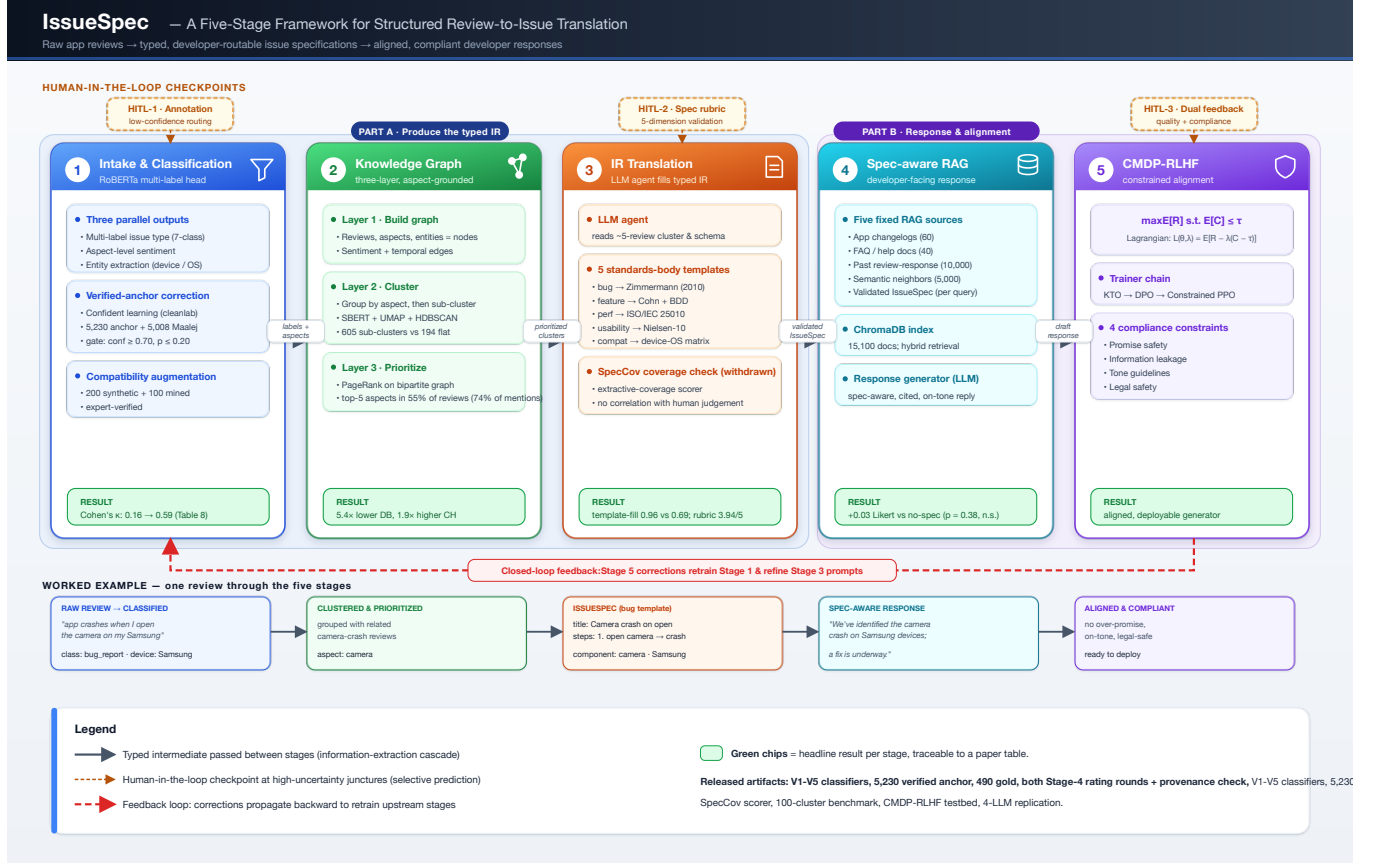


Figure 1: The IssueSpec five-stage pipeline. Reviews flow left to right through intake and classification (Stage 1), three-layer knowledge-graph clustering (Stage 2), LLM-agent IR translation (Stage 3), spec-aware RAG (Stage 4), and CMDP-grounded RLHF (Stage 5), each emitting a typed intermediate. Yellow checkpoints mark human-in-the-loop review under selective prediction [5, 24]; the red loop propagates Stage 5 corrections upstream.

2.5 RLHF and Constrained Optimization for Response Generation

DPO [50], KTO [20], and PPO [55] compress quality and compliance signals into a single scalar reward. Altman’s CMDP framework [1] separates these objectives, and Safe-RLHF [14] applies CMDP to general LLM safety with a domain-agnostic compliance set covering harmful, dishonest, and biased outputs. *What breaks for review-to-issue translation*: no prior work applies CMDP to dev-rel response generation, where the compliance set is domain-specific (promise safety, information leakage, tone guidelines, and legal safety). Our Stage 5 formulation differs by instantiating these four domain-specific constraints inside the CMDP Lagrangian and releasing a proof-of-concept training loop.

2.6 Cross-Category Positioning

Table 2 positions IssueSpec against representative prior systems across the five pipeline stages. Existing systems occupy at most two of the five stages; composing all five with a standardized issue schema, three theoretically-grounded HITL checkpoints, KG-centrality prioritization, and a CMDP-grounded RLHF formulation is the structural contribution of this work.

3 The IssueSpec Pipeline

The IssueSpec pipeline (Fig. 1) is a five-stage information-extraction cascade [28, 54]. Stages 1–3 produce the typed IR from raw reviews. Stage 4 uses it to drive spec-aware response generation. Stage 5 aligns the response generator via constrained RLHF. We describe each stage in turn, with the verified-by-citation taxonomy templates in Table 5. Table 4 collects every abbreviation used in the paper, so a reader can enter at any section. This section describes the framework only. Evaluation design, baselines, and metrics are in Section 4, exact settings and hyperparameters are in Appendix A, and results are in Section 5.

Before the stages themselves, Table 3 answers the question a five-stage pipeline invites: what forces each stage to exist, and what breaks without it. We answer it with the ablation that removes each one, so the justification is measured rather than asserted, and two of the five turn out not to justify themselves.

3.1 Producing the IR

Stage 1: Intake, Classification, and Aspect Extraction. Stage 1 produces three outputs per review. The first is a multi-label class from a RoBERTa head fine-tuned on Maalej [41] and Guzman [27]. Labels are multi-label because reviews mix concerns, as when a

Table 2: Cross-category positioning. Existing systems compose at most two of the five pipeline stages. IssueSpec is the first to compose all five with a standardized issue schema, three HITL checkpoints grounded in selective prediction theory, KG-centrality prioritization, and a CMDP-grounded RLHF formulation. “–” denotes a stage absent from that system.

System	Classify	Cluster	IssueSpec	Response	RLHF
AR-Miner [10]	filter	LDA topic	–	–	–
CLAP [60]	rating	DBSCAN	–	–	–
SURF [18]	intention	LDA	–	summary	–
RRGen [22]	–	–	–	seq2seq	–
CoRe [23]	–	–	–	context-aware retrieval	–
Maalej [41]	7-class	–	–	–	–
ReviewGraph [16]	rating	KG	–	–	–
SWE-Agent [62]	–	–	consumes (not produces)	–	–
Safe-RLHF [14]	–	–	–	general	dual-CMDP (general)
IssueSpec (ours)	V5 corrected	KG + PageRank	type-routed schema	spec-aware RAG	dual-CMDP (dev-rel)

Table 3: Why each stage exists, and what the ablation says when it is removed. Two stages do not justify themselves on our own evidence, and the paper says so rather than defending them.

stage	the problem it must solve	why it cannot be skipped
1 Intake	the corpus label is wrong 25% of the time, concentrated on minority classes	Stage 3 routes on the issue type, so a wrong type produces the wrong template
2 KG	215,583 reviews cannot be triaged one at a time; developers need an order	a spec is written per cluster, so reviews must be grouped and ranked
3 IR	trackers need typed fields, not prose	this is where the typed record is produced
4 RAG	a reply needs house voice and precedent, not just the defect	supplies phrasing the spec does not carry
5 CMDP	replies must not over-promise or leak internals	compliance is a hard constraint, not a reward term

Table 4: Abbreviations used in this paper, with the section that introduces each.

term	expansion
IR	Intermediate Representation, the typed issue spec
IssueSpec	one filled instance of the typed IR
KG	Knowledge Graph, the three-layer review index
HITL	Human-In-The-Loop checkpoint (HITL-1 to HITL-3)
SPECov	extractive-coverage score for a spec (withdrawn as a faithfulness measure, §5.1)
RAG	Retrieval-Augmented Generation
RLHF	Reinforcement Learning from Human Feedback
CMDP	Constrained Markov Decision Process
KTO	Kahneman-Tversky Optimization, a preference trainer
DPO	Direct Preference Optimization
PPO	Proximal Policy Optimization; CPPO is its constrained form
SFT	Supervised Fine-Tuning, the RLHF starting policy
NFR	Non-Functional Requirement (ISO/IEC 25010)
DB / CH	Davies-Bouldin / Calinski-Harabasz cluster indices

bug report also asks for a feature. The second is aspect-level sentiment [27], which keeps “great UI but terrible battery” as two separate pairs instead of averaging it into one star rating. The third is entity extraction for devices, OS versions, and named features. Stage 3 needs those entities as typed fields for the device-OS and bug-report templates. Low-confidence reviews route to HITL-1 under selective-prediction theory [24]: when the classifier cannot decide, the expert does, and the correction feeds the next training round.

LLM annotation noise is the main challenge here. On the 5,041-review praise stratum, 25% of LLM labels are wrong, concentrated on minority classes. Discarding them wastes annotation; using them as-is poisons training. We apply confident learning [46] against a verified anchor of 5,230 expert-labeled reviews plus 5,008

from Maalej [41], gated by a confidence threshold on the anchor and a probability ceiling on the LLM label (Appendix A); this keeps the good labels and corrects only the wrong ones. The compatibility class needs a separate fix: only 8 of 215,583 LLM labels are tagged compatible, and Maalej has none. We seed it with template-generated synthetic reviews [33] and keyword-mined RRGGen reviews, each checked by the lead author before training; counts are in Appendix A.

Stage 2: Three-Layer Knowledge Graph. Stage 2 organizes reviews into an aspect-grounded knowledge graph: build, cluster, rank. The first layer builds the graph: each review is a node linked to its aspects and entities by sentiment and temporal edges. A graph beats a flat list because reviews share aspects across users, and that co-occurrence is what later layers cluster and rank on. Aspects come from dependency-parse noun phrases following Hu and Liu [30], pruned by frequency and centrality to drop one-off terms.

The second layer clusters reviews first by aspect, then sub-clusters within each aspect using sentence embeddings [51], UMAP [44], and HDBSCAN [43]. Aspect-first keeps each sub-cluster on one theme: a flat clusterer mixes “battery on Samsung” with “battery in dark mode” because both share lexical content. The result is 605 aspect-grounded sub-clusters with mean size 3.0, against 194 from a flat baseline. The third layer prioritizes via PageRank [7, 47] on the bipartite review-aspect graph, ranking aspects by cross-review prevalence so developers see the most-affected first. Each cluster maps to the fixed schema in Table 5 with a centrality-derived priority score. The resulting aspect ranking is reported in Section 5.4.

Stage 3: Review-to-Issue Translation. Stage 3 instantiates the IssueSpec. An LLM agent reads the standardized cluster schema and writes a GitHub-issue-quality specification using the type-routed templates in Table 5. We use an LLM rather than rule-based extraction because reviews vary too much for fixed rules and the agent has to reason across a 5-review cluster. We route by issue type because a bug report needs reproduction steps, a feature request needs acceptance criteria, and a performance complaint needs latency or memory numbers; a shared template would be incomplete or padded. The five templates are Zimmermann bug reports [65], Cohn user stories [12] with BDD acceptance criteria [61], ISO/IEC 25010 non-functional categories [21], Nielsen heuristics [45], and device-OS matrices [33]. Each is grounded in its standards-body citation because developers already recognize these formats.

The output passes through HITL-2: a domain expert scores each spec on a five-dimension rubric (completeness, accuracy, actionability, specificity, clarity, each 1–5), following the LLM-as-judge calibration convention of Gilardi et al. [25], Pangakis et al. [48], and Zheng et al. [64]. Five dimensions, not one, because a single quality number conflates completeness with clarity and the LLM cannot tell which one to fix on retry. Any spec failing on any dimension goes back to the LLM with dimension-level feedback for a second pass.

SPECcov faithfulness metric. We release SPECcov, an extractive-coverage scorer that measures substantive-token overlap (length ≥ 5 , stop-word filtered) with quoted-string and coverage bonuses. Bands follow news-summarization conventions [26]; the metric targets the extractive-coverage sub-type of faithfulness [26, 42], a known lexical correlate of factual consistency [37] (the remaining hallucination sub-types in Ji et al. [31] are out of scope). Scores are reported with the other Stage 3 results in Section 5.1. **What SPECcov is and is not.** It is a lexical-grounding proxy in the sense of [26]: it measures whether the content of a spec can be traced back to tokens in its source cluster. It is not a faithfulness classifier, and Section 5.1 reports the experiment that establishes this. We describe it here because it is part of the released pipeline and because the negative result only makes sense against its definition.

3.2 Using the IR in Response Generation

Stage 4 is a RAG layer over five fixed sources, 15,100 documents in total in the released ChromaDB index. Four of them are pre-indexed: app changelogs and release notes (60 docs), FAQ and help documentation (40), 10,000 past review-response pairs from RRGGen [22], and 5,000 semantically similar past responses. The fifth is the validated IssueSpec from Stage 3, injected per query rather than pre-indexed, because specs are generated per cluster. These five were chosen to give the generator the answer (changelogs, FAQ), the voice (past responses), the closest precedent (semantic neighbors), and the structured intent (IssueSpec) without overlap.

3.3 Aligning the IR-Aware Generator

Stage 5 frames alignment as a CMDP [1, 14]: maximize a quality reward subject to a compliance constraint,

$$\max_{\theta} \mathbb{E}_{\pi_{\theta}} [R_{\text{quality}}] \quad \text{s.t.} \quad \mathbb{E}_{\pi_{\theta}} [C_{\text{compliance}}] \leq \tau,$$

with the Lagrangian relaxation

$$\mathcal{L}(\theta, \lambda) = \mathbb{E}_{\pi_{\theta}} [R - \lambda(C - \tau)]$$

collapsing gracefully to single-objective optimization whenever the constraint is already satisfied. We use a CMDP rather than a scalar reward because a dev-rel response can be high-helpfulness and still out of compliance, like a polite promise the team cannot deliver or a leaked roadmap detail, and a weighted sum hides exactly the trade-off the constraint is meant to expose. The compliance scorer tracks four domain-specific dimensions absent from prior general-safety RLHF [14]: promise safety (no commitments the team cannot keep), information leakage, tone guidelines, and

legal safety. The trainer is Constrained PPO [14, 55] with a KL-regularized reference policy that keeps updates from drifting too far from the supervised baseline. KTO [20] and DPO [50] are reported as small-scale single-objective comparators in Section 5.3; together they bracket the preference-data spectrum, KTO needing only binary feedback and DPO needing pairwise.

HITL-3 supplies two parallel feedback streams. *Quality* feedback (helpfulness, specificity, empathy, accuracy, actionability) is the reward; *compliance* feedback (promise safety, leakage, tone, legal) is the constraint. Feedback also propagates backward: classification corrections retrain Stage 1, rubric scores refine Stage 3 prompts, and response scores drive the RLHF updates, closing the Stage 5→Stage 1 loop in Fig. 1.

4 Experimental Setup

4.1 Dataset, Deduplication, and Annotation Provenance

We use the RRGGen corpus [22]: 310,031 review-response pairs from 58 Android applications. After exact-text deduplication that removes 94,448 duplicate reviews (−30.5% of the raw corpus), 215,583 unique reviews form the working corpus, labeled across the seven Maalej [41] classes. Evaluation additionally uses a 490-review expert gold standard (70 per class $\times$ 7 classes) and 100 stratified clusters for Stage 3 evaluation.

Annotation provenance (full corpus, 215,583 reviews). Of the working corpus, 79.49% of labels come from the V2 LLM annotator with no override, 18.08% are cleanlab-corrected against the verified anchor, and 2.43% are directly human-verified by the lead author. The lead-author-versus-V2-LLM deviation is 25%, replicated independently across two strata: the 5,041-review praise stratum and the 5,230-row human-verified anchor subset (Table 6). This $\approx 25\%$ deviation rate, an order of magnitude higher than i.i.d. label-noise assumptions, motivates the confident-learning procedure of Stage 1.

4.2 Annotation Methodology

Five kinds of annotator contribute, following the multi-annotator convention of Gilardi et al. [25] and Pangakis et al. [48] who treat LLM annotators as independent raters for inter-rater statistics.

(i) Lead author (domain expert). Verified 5,230 reviews across minority strata (approximately thirty person-hours), wrote 20 reference IssueSpecs, labeled the 490-review classification gold blinded to V2 LLM, and scored the 400-row Stage 4 set on quality (1–5), specificity (1–5), and helpful (Y/N) with method IDs hidden until aggregation.

Label provenance, stated plainly. No LLM-generated label is used as ground truth anywhere in this paper. LLM labels are the *object under correction*, not the yardstick. The V2 corpus labels are treated as noisy input, the confident-learning step proposes corrections against a human-verified anchor, and every reported accuracy or κ is measured against a gold standard produced by human annotators. Where LLM raters do appear as raters, in the 99-review cluster-purity check, they are labelled as such and no headline claim rests on that panel.

How the gold was constructed, and what that costs. Annotators did not label freely. Each was shown the V5 classifier’s

Table 5: IssueSpec standardized issue schema. Each issue type follows the dominant industry or HCI standard; templates are verified against their standards-body citations, producing GitHub-issue-quality artifacts a defect tracker can consume directly.

Issue type	Template (citation)	Justification
bug_report	Zimmermann [65]: title, severity, steps_to_reproduce, expected, actual, component	Standard in GitHub Issues, JIRA, defect trackers; Chaparro et al. [9] confirm these are the most-requested bug-report fields
feature_request	Cohn user-story [12] (As-a / I-want / So-that) + behavior-driven development (BDD) acceptance criteria [61]	Dominant industry format (Agile, Scrum, SAFe); BDD standardizes testable outcomes
performance	ISO/IEC 25010 [21] performance-efficiency sub-characteristics: time behavior (speed, responsiveness), resource utilization (memory, battery), capacity (scalability)	International standard for software-quality NFR taxonomy; the parenthesized terms map ISO sub-characteristics to mobile-app vocabulary
usability	Nielsen-10 heuristics [45] (visibility, match-real-world, user control, etc.)	Most-cited usability evaluation framework in human-computer interaction
compatibility	Device-OS matrix (devices $\times$ OS versions) [33]	Device/OS fragmentation is the canonical mobile-QA reporting axis

predicted label and asked whether it was correct, supplying a replacement when it was not, and the written protocol told them to accept a reasonable label and overturn only when confident. This makes annotation fast and it makes agreement high, and both effects flatter us. It also couples the yardstick to the system being measured: V5’s label is the starting point for every item, and V2 and the cleanlab-corrected labels never received that benefit of the doubt. The sample is stratified over V5’s predicted classes as well, which forces 70 items per class into a set where V2 predicts two of those classes almost never, so V2’s per-class F1 of 0.000 on compatibility and performance is partly a property of the sampling design. The κ trajectory in Table 12 should be read as a comparison on a V5-seeded, class-balanced sample rather than as a neutral benchmark, and a free-labelling gold would be the way to settle it.

How rater conflicts are resolved. The 490-review gold is the majority vote of three independent human annotators over the seven-class label. Of the 489 usable items, 476 (97.3%) are unanimous and 13 (2.7%) are two-to-one splits resolved by the majority; there are no three-way ties, so no item required a tie-break rule or a further adjudication round. One item was marked wrong by all three raters without a replacement label supplied and is excluded rather than guessed. Disagreement is not discarded silently: the per-item votes are released with the gold standard, so a reader can recompute any statistic on the unanimous subset alone. Before the main task all three annotators completed a shared 20-review calibration round whose disagreements were discussed and resolved against the written decision rules, which is where the systematic conflicts (slow versus broken, device-specific versus general) were settled rather than in the main pass.

A disclosure about an earlier annotation round. The 490-review gold was annotated twice. In the first round three sheets were distributed, and the two returned alongside the lead author’s carried annotation columns identical to hers cell for cell, so no genuine agreement could be computed from them and they are not used anywhere in this paper. The round we report is the second, described below, and the agreement statistics come only from it. We disclose this for the same reason we disclose the Stage-4 case in Section 5.2: a paper that offers a check for non-independent ratings should say where its own artifact failed that check. Both rounds are released.

(ii) Independent human rater panel (three raters). The 490-review classification gold is annotated independently by the lead author and two further human raters on the identical item set. Each

rater sees the classifier’s predicted label and marks it correct (Y) or wrong (N), supplying the correct class when N. All three complete a shared 20-review calibration round with adjudicated disagreements before the main task opens, and work from the same written decision rules (Appendix B). The gold label for each review is the majority vote over the three raters. This is a verification task rather than open labeling, which is why agreement is high relative to free-labeling studies of app reviews.

(iii) Stage 4 rater panel (same three people). Stage 4 was rated twice. The first round scored a 400-row set whose conditions were produced by template composers; we discarded it, for reasons given in Section 5.2. The round we report is a 200-row set in which both arms come from the same model and differ only in whether the IssueSpec was supplied. All three raters scored all 200 rows independently, blind to the condition behind each reply, and we release both rounds together with the independence check that separates them. Three human annotators contribute in total, the lead author and two collaborators; categories (iv) and (v) are model raters, not people.

(iv) LLM-as-judge (Qwen2.5-3B-Instruct). Scored 28 IssueSpecs on the five-dimension rubric and 50 sub-clusters on the Y/P/N purity rubric [64], both calibrated against the lead author.

(v) Corpus-labeling LLMs. The V2, cleanlab-V2, and V5 LLMs are treated as independent raters for inter-rater statistics.

Table 6 consolidates agreement and disagreement across all pairs.

Ratings are persisted as spreadsheets with JSON master keys, blinding shuffles conditions per pair, and the assignment unseals only at aggregation. Two distinct three-rater statistics appear in Table 6 and should not be conflated. The classification gold uses three independent *human* raters on 490 reviews and reaches Fleiss $\kappa = 0.968$ on the Y/N verdict and 0.979 on the effective seven-class label, with no three-way ties. The $\alpha = 0.451$ row is a separate cluster-purity check on 99 reviews that uses the lead author plus two Qwen2.5-3B prompts (concise and chain-of-thought) as raters.

4.3 Multi-LLM Bias Control

A single-LLM headline raises a structural-bias concern: results from any one model may not transfer across families. We therefore replicate Stage 3 across four LLMs drawn from three distinct families, Anthropic Claude (Opus 4.7), Meta Llama-3.3-70B-Instruct (via Groq), and Alibaba Qwen2.5-{3B, 1.5B}-Instruct, with Stage 4

Table 6: Annotator-vs-LLM and inter-annotator deviation. The $\approx 25\%$ deviation is replicated on two independent strata; the κ -progression reaches moderate agreement [39]; the V5 endorsement row is a consistency check, not independent validation, since V5 trained on those corrections; LLM-judge calibration Δ bounds judge bias.

Annotator pair	sample	agreement / deviation
Lead-author vs V2 LLM	10,271	75% / 25% mislabeled
Lead vs V2 / cleanlab / V5	490 gold	κ : 0.163 $\rightarrow$ 0.333 $\rightarrow$ 0.592
3 human raters, Y/N verdict	490 gold	Fleiss $\kappa = 0.968$; 479/490 unanimous
3 human raters, 7-class label	489 gold	Fleiss $\kappa = 0.979$; $\alpha = 0.979$
V2 LLM vs V5; cleanlab vs V5	215,583	71.96% / 86.77% agree
cleanlab vs V5 (consistency, not independent)	40,291	88.66% support
V5 vs gold, strictly held out	307	$\kappa = 0.616$
3 human raters (Stage 4 quality)	400 rated rows	weighted κ 0.970–0.986; within-1 100%
3 human raters (Stage 4 helpful)	400 rated rows	κ 0.678–0.836; $\alpha = 0.781$
3-rater LLM-assisted purity check (lead + 2 Qwen prompts)	99 reviews	Krippendorff $\alpha = 0.451$
Lead vs Qwen-judge (purity, rubric)	50 / 28	$\Delta = -0.035 / +0.23$

replicated on a Claude-vs-Qwen-3B subset, and check that cross-family rank order is preserved (Table 9). Frontier-family replication on GPT-4o and Gemini-1.5-Pro is queued.

4.4 Experimental Design

We pre-register three experiments, one per research question:

Experiment 1 (RQ1, translation quality). Five conditions are compared on the same 100 stratified clusters: (a) LLM with taxonomy grounding (the full Stage 3 system), (b) LLM without taxonomy grounding (free-form), (c) raw review summary (a lower bound that concatenates the cluster’s reviews), (d) lead-author reference, and (e) human-written GitHub issues mined from three open-source Android projects (the practitioner-quality upper bound). Each spec is scored on the five-dimension rubric. Significance is reported via paired Wilcoxon signed-rank tests on rubric scores between conditions.

Experiment 2 (RQ2, coupled vs. uncoupled), as reported. The design we report holds the generator fixed. The same model, given the same retrieved context, writes two replies for each of 100 reviews, differing only in whether the validated IssueSpec is in the prompt. All 200 rows are rated blind by three raters on quality, specificity, and helpfulness, and analysed with a paired Wilcoxon test and a paired bootstrap interval. *The discarded design.* An earlier round used four conditions produced by template composers rather than by a model; we describe it because the paper reports its withdrawal, not because it supports a claim.

Experiment 2, discarded design. The same 100 clusters drive four response-generation conditions: (a) an rrgen-style baseline (review only), (b) a prompt-baseline (CoRe-style with dev-rel prompt and lightweight entity extraction), (c) RAG without IssueSpec, and (d) the full system with IssueSpec. Evaluation uses automatic metrics (BLEU, ROUGE-L, BERTScore) and 400 blinded paired human ratings (quality 1–5, specificity 1–5, helpful Y/N), with the lead author as primary annotator and a second rater on a 30-pair subset. Statistical significance is checked simultaneously by paired Wilcoxon, Friedman with Nemenyi post-hoc [17],

Bradley-Terry, and McNemar at Bonferroni-corrected $\alpha = 0.05$. Effect sizes follow Romano et al. [52] and Cliff [11].

Why these four conditions. The Experiment 2 ladder is designed so each rung adds exactly one thing. The prompt baseline adds dev-rel system instructions on top of an rrgen-style review-only baseline, no_spec then adds retrieval, and full then adds the validated IssueSpec. Reading the ladder from bottom to top isolates which single addition moves human-rated quality. The headline comparison runs single-shot vanilla RAG [40] rather than an agentic loop, so that the IssueSpec’s contribution is not confounded with orchestration; the agentic-RAG comparison against Self-RAG [4] and DSP [36] runs separately on a 10-review feasibility sample (stratified, two per issue type, Qwen2.5-3B as critic and reviser, up to two iterations) and is reported as a demonstration rather than a powered comparison.

Experiment 3 (RQ3, dual- vs. single-objective RLHF). Five policies run on a distilGPT2 proof-of-concept (82M parameters, 400 SFT samples, 296 KTO pairs, 100 DPO pairs): SFT-base, single-objective KTO [20], single-objective DPO [50], a constrained-proxy variant, and dual-objective Lagrangian Constrained PPO [14, 55]. We measure head-to-head BLEU, ROUGE-L, and BERTScore on a 100-review test set, plus the rule-based safety-violation rate.

5 Results

How to read this section. Each research question owns one subsection, one headline figure, and one headline table, and nothing else in the section is required to follow the argument. Section 5.1 answers RQ1, is the typed IR producible at scale, with Table 11 for template-fill and Figure 2 for the classifier recovery that makes corpus-scale production possible. Section 5.2 answers RQ2, is the IR load-bearing downstream, with Table 14; the answer is no, and the subsection explains what our earlier positive answer was measuring. Section 5.3 answers RQ3, does constrained alignment demonstrate a quality-versus-compliance trade-off, and reports a negative result. Sections 5.4 and 5.5 are supporting evidence: they test whether individual components earn their place, and Section 6.1 states plainly which ones do. A reader tracking only the contribution can read Sections 5.1, 5.2, and 6.1.

Table 7 states what we take away from the section before the evidence for it, so a reader can decide which subsections to read.

5.1 Is the IR Producible at Scale?

Template-fill measures schema conformance, and we state its limit before the number. The criterion is the templates’ own required fields, so a generator that was never asked for a reproduction-step list cannot score on one; the comparison shows that routing produces the fields, not that routed specs are better written. Under strict content-validity criteria (at least three reproduction steps with action verbs, descriptions ≥ 30 words, and a full As-a/I-want/So-that triple for user stories), the LLM-with-taxonomy condition achieves a substantive template-fill rate of 0.96 (Table 11). Free-form LLM reaches 0.69, the same as the lead-author reference set ($n = 20$). Raw concatenation lands at 0.34, and real GitHub issues mined from three open-source Android projects reach 0.53. The LLM-vs-GitHub gap reflects coverage rather than

Table 7: What the experiments establish, in one line each. Negative findings are marked, and they are findings rather than caveats.

	finding	evidence
F1	Type routing, not the language model, is what produces tracker-ready fields: removing it collapses strict fill from 0.96 to 0.69 and the per-type structural metrics to zero.	\$5.1, A3
F2	LLM annotation noise on this corpus is structurally biased toward minority classes, not spread uniformly, which is why correcting against a small verified anchor works at all.	\$5.1
F3	<i>Negative, and it overturns our own earlier claim.</i> With the generator held fixed, adding the typed IR changes response quality by +0.04 ($p = 0.38$); the +2.35 we previously reported was an artifact of comparing two different text composers.	\$5.2
F4	Generator capability shows up mostly as whether fields get filled, not as the quality of the text once filled: template-fill spans 0.96–0.48 across four models while rubric quality spans 4.09–3.77.	\$5.1
F5	<i>Negative.</i> The knowledge graph loses to count-matched flat clustering on all three intrinsic metrics; its defensible role is centrality-based prioritization.	\$5.4, A3
F6	<i>Negative.</i> Extractive coverage does not measure faithfulness: SPECov shows no rank correlation with human judgement and orders generators backwards.	\$5.1
F7	<i>Negative, then partial.</i> A CMDP constraint cannot be tested on a policy incapable of violating it; with one that can, the constraint binds and λ grows, but no trade-off is demonstrated.	\$5.3
F8	Run-to-run generation noise is about 0.03 on template-fill, which is the yardstick separating the differences we claim from the ones we do not.	\$6.4

content quality: the LLM is prompted to fill every field, whereas humans leave optional fields blank. A very-strict sensitivity sweep preserves the ordering, Claude drops from 0.97 to 0.71, with the Claude-vs-GitHub gap shifting only -0.085 .

SPECov scores, and a correction to how they were computed. An earlier version of our scorer applied per-condition floors, raising `raw_summary` to 5 and the human reference to 4 regardless of content, on the reasoning that verbatim copying is faithful by construction and hand-written references deserve a floor. That is not a measurement, because a score whose value depends on the label of the thing being scored cannot compare labels, so we removed the floors and report what the scorer produces unaided. Across 320 specs the LLM-with-taxonomy condition averages 4.19, free-form LLM 3.38, raw concatenation 4.47, and the human reference 3.45. The two LLM conditions are unchanged, because no floor was ever applied to them; the floors had inflated the two comparison conditions, so removing them makes our own system look better rather than worse, and we note that explicitly so the direction of the correction is on the record. Read on their own these numbers would say the taxonomy condition scores highest. Section 5.1 shows they should not be read on their own. Top-ranked aspects by PageRank centrality are `ad` (0.040), `phone` (0.011), and `battery` (0.007).

SpecCov does not measure faithfulness. An extractive-coverage score is only useful if it agrees with what a person calls faithful. We tested that directly. Three human raters independently scored 60 specs, 20 from each of the three generators, shuffled and unlabelled, on a 1–5 faithfulness scale, and separately listed every concrete claim the source reviews do not support. One qualification is needed before the numbers. One rater completed the corrected sheet, which shows exactly the evidence SPECov itself scores against; the other two completed an earlier draft that showed a subset of it, so they judged against less evidence than the metric uses and their flags are biased toward over-flagging. We therefore

lead with the rater who saw the full evidence and report the other two beside her. The metric fails all three checks on either reading. *It does not rank specs the way humans do.* Spearman ρ between SPECov and human faithfulness is +0.045 ($p = 0.73$), +0.015 ($p = 0.91$), and +0.226 ($p = 0.08$) for the three raters, and +0.147 ($p = 0.26$) against their mean. No correlation reaches significance.

It does not detect the claims humans call unsupported. On the 60 specs, one rater marked 45 as containing at least one unsupported detail and 15 as clean. SPECov scores those two groups at 3.64 and 3.93, a difference in the right direction but far from significant (Mann-Whitney $p = 0.34$). A metric that detected hallucination would separate these groups.

It orders the generators backwards. SPECov ranks the taxonomy condition highest (4.20) and the human reference lowest (3.45); the humans rank them the other way, human reference 4.10 and taxonomy 3.50. The mechanism is visible in the definition: templated specs reuse source vocabulary densely, which is what the score rewards, while a human writer condenses and paraphrases, which the score penalizes even when nothing is invented.

We therefore withdraw SPECov as a faithfulness measure. Lexical grounding is not a usable proxy for faithfulness on this task, and the automatic 320-spec rubric that appeared to support it is not independent evidence, since its faithfulness dimension is computed by the same overlap procedure. Entailment-based measurement [19, 29, 38] is the direction we would now take. We release the scorer, the 60-spec human ratings, and this analysis, because a documented negative result on a plausible-looking metric is more useful to the next person than a metric that was never tested.

Cross-LLM replication (bias control). Table 9 reports four uses across three model families. Llama-3.3-70B reaches 0.80 template-fill, Qwen2.5-3B 0.74, and Qwen2.5-1.5B 0.48 under the same strict criteria; the templated > free-form > raw ordering holds across all four models with capability-scaled magnitudes. On two per-class structural-compliance checks, Llama-3.3-70B attains complete template compliance on the stratified sample: all 29 bug specs include three or more reproduction steps with action verbs (vs. Claude’s 73%), and all 30 feature specs include the *As-a/I-want/So-that* triple (vs. Claude’s 97%). These measure structural template compliance (does the field exist with the required form?), not content correctness, content quality is reported separately under the five-dimension rubric. The structural-bias control that matters here is cross-family rank preservation: the IssueSpec contract transports across families rather than depending on a single proprietary generator.

Cross-family rubric quality. Template-fill measures whether a generator emits the required fields; it says nothing about whether the content is any good. We therefore ran the same Qwen2.5-3B judge and the same five-dimension rubric over the other generators’ specs, on matched clusters (Table 8). On the 14 clusters all four generators cover, mean rubric score follows the same order as template-fill: Claude 4.09, Llama-3.3-70B 3.96, Qwen2.5-3B 3.83, Qwen2.5-1.5B 3.77. We state plainly what that ordering is worth. At $n = 14$ none of the six pairwise differences reaches significance (Wilcoxon p between 0.18 and 0.75), and the adjacent gaps are the same size as the run-to-run generation noise measured in Section 6.4. The four-way ordering is therefore consistent with the template-fill result rather than independent confirmation of it. The

Table 8: Cross-family five-dimension rubric, same Qwen2.5-3B judge as Table 10, on matched clusters. Panel A is the four-way comparison on clusters covered by every generator (bug reports only). Panel B widens the Claude-versus-Llama comparison to all five issue types. The Panel-A ordering matches the template-fill ordering in Table 9, but none of its six pairwise differences is significant at $n = 14$ (Wilcoxon p from 0.18 to 0.75), so read it as descriptive. Panel B, with twice the sample and all five issue types, does separate: +0.38 for Claude over Llama, Wilcoxon $p = 0.015$.

generator	n	mean	compl.	accur.	action.	spec.	clarity
<i>Panel A, four-way, 14 matched bug-report clusters</i>							
Claude (Anthropic)	14	4.09	4.00	4.50	3.71	4.36	3.86
Llama-3.3-70B (Meta)	14	3.96	3.86	4.50	4.07	3.86	3.50
Qwen2.5-3B (Alibaba)	14	3.83	3.07	4.50	4.00	3.86	3.71
Qwen2.5-1.5B (Alibaba)	13	3.77	3.00	4.15	4.46	3.62	3.62
<i>Panel B, Claude vs Llama, 28 clusters, all five issue types</i>							
Claude (Anthropic)	28	4.01	3.86	4.54	3.75	4.29	3.64
Llama-3.3-70B (Meta)	28	3.64	3.39	4.46	3.14	3.82	3.36

Table 9: Multi-LLM evidence summary across three model families (Anthropic, Meta, Alibaba). Cross-family rank preservation across Stage 3 and Stage 4 is the structural-bias control; LLM-as-judge calibration Δ vs the lead author bounds judge bias.

Use	Model (family)	Result
Stage 3 generate	Claude (Anthropic) / Llama-3.3-70B (Meta) / Qwen-3B (Alibaba) / Qwen-1.5B (Alibaba)	0.96 / 0.80 / 0.74 / 0.48
Stage 4 generate	Claude / Qwen-3B	0.61 / 0.54
5-dim rubric judge	Qwen-3B, all 100 specs	3.94/5 (28-spec subsample: 3.89)
Cross-family rubric	Claude / Llama / Qwen-3B / Qwen-1.5B	4.09 / 3.96 / 3.83 / 3.77 (Table 8)
Y/P/N purity judge	Qwen-3B vs lead-author	0.625 vs 0.66; $\Delta = -0.035$

Claude-versus-Llama comparison in Panel B, which has twice the sample and spans all five issue types, does reach significance at +0.38 ($p = 0.015$), and that is the one cross-family quality difference we assert. The spread is much narrower than the template-fill spread (0.96 to 0.48), which is informative in itself: the capability gap between these models shows up mostly as whether fields get filled at all, not as the quality of the text once they are. Completeness is the dimension that separates them (4.00 down to 3.00), while accuracy is nearly flat (4.50 for three of the four). On the wider Claude-versus-Llama panel, which spans all five issue types, Claude leads 4.01 to 3.64, with the gap concentrated in actionability and completeness. Two caveats belong with these numbers. The four-way panel is bug-report clusters only, because that is where the two Qwen runs overlap, and the judge is itself Qwen2.5-3B-Instruct, so the Qwen2.5-3B row carries a self-preference risk that the others do not.

Five-Dimension Rubric. The proposal specifies a 5-dimension Likert rubric (completeness, accuracy, actionability, specificity, clarity, each 1–5) [25, 48, 64], operationalized with Qwen2.5-3B-Instruct as the judge. We score the complete set rather than a subsample: all 100 taxonomy specs and all 20 lead-author reference specs, in a single run (Table 10). The taxonomy condition

Table 10: Experiment 1 five-dimension rubric (Qwen-judge, 1–5 scale) on the *complete* spec set: all 100 taxonomy specs and all 20 lead-author reference specs, scored in one run so the two columns are directly comparable. The middle column repeats the earlier stratified 28-spec subsample under the same judge and rubric; the gap between it and the full set is +0.05, which is the subsample error. The LLM-as-judge protocol follows the calibration-against-reference convention of [25, 48, 64].

dimension	taxonomy, $n = 100$	taxonomy, $n = 28$	lead-author ref, $n = 20$
completeness	3.72	3.79	3.20
accuracy	4.43	4.39	4.10
actionability	3.86	3.71	2.85
specificity	4.15	4.07	4.05
clarity	3.52	3.50	3.05
overall mean	3.94	3.89	3.45
sd across specs	0.59	–	0.41

Table 11: Experiment 1 substantive template-fill (strict criteria). The 0.96 vs 0.53 LLM-vs-GitHub gap is coverage, not content quality; the strict-vs-very-strict sweep preserves ordering. The exact-zero cells in bugs:repro and features:story columns for (b), (c), (d), and (e) are structural: free-form LLM, raw concatenation, lead-author, and human-GitHub specs do not emit the template-specific reproduction-step list or As-a/I-want/So-that triple at all when not prompted for them, so the strict-fill rate is zero by construction.

condition	strict	bugs:repro	features:story
(a) LLM + taxonomy	0.96	0.73	0.97
(b) LLM free-form	0.69	0.00	0.00
(c) raw concat	0.34	0.00	0.00
(d) lead-author ref ($n = 20$)	0.69	0.00	0.00
(e) human GitHub (3 repos)	0.53	0.13	0.00

reaches 3.94/5 (sd 0.59 across specs), above the hypothesized lower bound of 3.5. An earlier stratified 28-spec subsample under the same judge and rubric gave 3.89, so subsampling cost 0.05 of a Likert point here, which bounds how much the small-sample version of this number could have been misleading. Per issue type the taxonomy condition ranges from 4.23 on performance and 4.02 on bug reports down to 3.45 on usability, the type whose reviews are least specific about what went wrong.

The reference set is a judge calibration, not a competitor. It scores 3.45 overall, below the generated specs, and the gap is concentrated in actionability (2.85 versus 3.86). The reason is visible in the sheets: the lead author filled title, description, severity, and component but left the template-specific fields such as reproduction steps and acceptance criteria empty, because writing a reference spec by hand is a different task from filling a routed template. The comparison therefore says that a human writing freely produces less tracker-ready output than a type-routed generator, which is the same effect the 0.96-versus-0.53 template-fill gap measures against real GitHub issues. It is not evidence that the specs are better than expert judgement on content, and we do not read it that way.

Stage 1 Classification Recovery. The Cohen’s κ trajectory against the 490-review expert gold crosses three Landis-Koch [39] agreement bands (Table 12): the V2 LLM achieves $\kappa = 0.163$ (slight);

after cleanlab correction [46] $\kappa = 0.333$ (fair); the V5 classifier retrained on the corrected labels reaches $\kappa = 0.592$ (moderate, just below the substantial threshold). The trajectory does not depend on any single annotator. Scored against the majority-vote gold of the three-rater human panel ($n = 489$ after excluding one review left without a correction label), the same three classifiers reach $\kappa = 0.165, 0.334$, and 0.590 , within 0.002 of the lead-author values at every step (Table 12). On the strictly held-out 307-review subset (183 of the 490 entered training via the verified anchor), V5 reaches $\kappa = 0.616$, slightly above the 0.592 on the full 490, ruling out a contamination artifact.

The per-class recovery story (Table 13) makes the structural collapse visible. On the two minority classes the V2 LLM cannot see at all, F1 jumps to 0.83 for compatibility and 0.77 for performance. Applied across the full 215,583-review corpus, V5 agrees with the cleanlab correction on 88.66% of the 40,291 corrected cases. We want to be exact about what that number is and is not, because it is easy to over-read. V5 was trained on a corpus in which 38,985 of those corrected rows carry the corrected label, so its agreement is substantially a measure of fit to its own training signal, not an independent second opinion. We report it as a consistency check, which is what it is, and we do not claim the two signals are independent.

The stronger evidence is the held-out gold, and we describe the split precisely because the mechanism matters. V5’s training corpus is built with a stratified per-class cap of 15,000 rows at seed 42. Replaying that cap identifies which corpus rows survived into training: 183 of the 490 gold reviews did, and 307 were capped out and never seen. The split is therefore an artifact of class-balancing rather than of anchor membership. That also limits what it can show, and we were wrong to describe it earlier as unbiased. Because the cap binds only on the large classes, the 307 unseen reviews contain only bug_report, praise, other, and feature_request; every performance, usability, and compatibility item sits in the seen 183. Those are the classes carrying the recovery, so the held-out comparison tests memorisation on the majority classes and says nothing about the minority-class gains. On the 307 unseen reviews V5 reaches $\kappa = 0.616$, slightly above the 0.592 it scores on the full 490. A model that had merely memorised its training labels would fall on unseen items rather than hold. One caveat belongs with this. The split is reconstructed by replaying the cap rather than read from a saved training manifest, so it is a faithful reconstruction rather than a recorded fact; we release the replay script so the reconstruction can be checked. The choice of confident-learning thresholds is not load-bearing: a full 3×3 grid over (anchor_conf, max_llm_prob) $\in \{0.65, 0.70, 0.75\} \times \{0.15, 0.20, 0.25\}$ keeps V5 endorsement in an 82.7–91.0% band, with the headline (0.70, 0.20) at 88.66% sitting near the grid center.

To rule out memorization of the 200 synthetic compatibility templates used during V5 training, we evaluate V5 against Maalej’s 5,008 labels under a label-space cross-protocol test. Review texts appear in the anchor under the RRGGen seven-class taxonomy, but predictions are scored against Maalej’s separate taxonomy, which contains no compatibility class, making any compatibility predictions out-of-distribution with respect to the evaluation labels. On the five overlapping classes ($n = 4,560$),

Table 12: Cohen’s κ progression vs the 490-review expert gold. Three Landis-Koch [39] bands crossed; cleanlab correction follows the confident-learning procedure of [46]. The final column repeats the measurement against the majority-vote gold of the three-rater human panel, showing the trajectory is not an artifact of a single annotator.

classifier	n	Cohen’s κ	Landis-Koch band	κ vs 3-rater gold
V2 LLM original	490	0.163	slight	0.165
cleanlab-corrected V2	490	0.333	fair	0.334
V5 classifier	490	0.592	moderate	0.590

Table 13: Per-class F1 against the 490-review expert gold. V2 LLM is blind to the two minority classes (compatibility, performance). The verified-anchor procedure with targeted augmentation recovers them.

class	V2 LLM	cleanlab-V2	V5
bug_report	0.43	0.50	0.71
feature_request	0.59	0.61	0.69
performance	0.00	0.47	0.77
usability	0.11	0.21	0.55
compatibility	0.00	0.00	0.83
praise	0.41	0.55	0.79
other	0.27	0.36	0.41
macro F1	0.26	0.39	0.68

V5 achieves macro F1 = 0.676, weighted F1 = 0.730, and accuracy = 72.0%. V5 also flags 432 reviews (8.6%) as compatibility; the top-twenty-confidence flags read as canonical compatibility complaints (“crashes on iphone 5 and ipad 3”, “works on HTC Desire X but not Samsung Galaxy Tab 3”), all labeled bug_report in Maalej because Maalej offers no alternative class. The detector is performing concept-recognition, not template-matching.

5.2 Is the IR Load-Bearing? A Result We Had to Withdraw

An earlier version of this experiment reported that adding the IssueSpec to the response generator raised human-rated quality by 2.35 Likert points. That number was wrong, not in its arithmetic but in what it measured, and the correction is the most important thing this section has to report.

What was wrong. The four conditions in that experiment were not produced by a language model. Each was produced by a deterministic template composer, and the composers differed in authorial register as much as in what information they were given. The no_spec arm emitted lemmatised, ungrammatical text averaging 78 words (“we be sorry that the login loop you describe have got in the way”); the full arm emitted polished prose averaging 123 words, with cleanup passes and an empathy phrasebook that no other arm received. A rater comparing them was separating two writing styles, and the presence or absence of the IssueSpec was confounded with that separation. The gap measured the composer at least as much as the specification.

The clean comparison. We re-ran the experiment with the confound removed. Both arms are now generated by the same model, given the same retrieved context, on the same 100 reviews, and differ in exactly one respect: whether the validated IssueSpec appears in the prompt. The two arms average 105 and 97 words

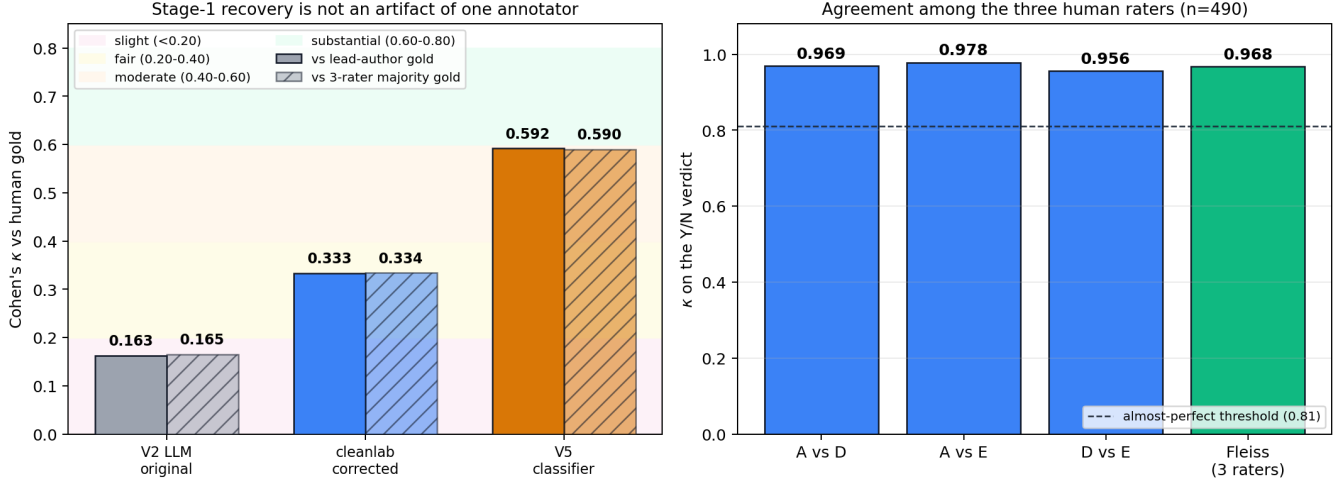


Figure 2: Stage-1 classifier recovery and the gold standard it is measured against. Left: Cohen’s κ for the three classifier versions, plotted twice, once against the lead-author gold (solid) and once against the three-rater majority-vote gold (hatched). The two are within 0.002 at every step, so the 0.163 to 0.592 trajectory does not depend on any single annotator. Background bands are the Landis-Koch [39] agreement ranges. **Right:** agreement among the three human raters on the same 490 reviews, pairwise and as a three-rater Fleiss κ , all above the almost-perfect threshold. Read the left panel for the result and the right panel for the reliability of the yardstick it uses.

with distinct-1 of 0.736 and 0.760, so they are not separable on surface features. Three raters scored all 200 blinded rows.

The IssueSpec does not measurably improve response quality. Pooled over the three raters the gain is +0.04 Likert points, 95% bootstrap CI $[-0.10, +0.18]$, paired Wilcoxon $p = 0.38$ on 100 pairs, Cliff’s $\delta = 0.04$. The blinded slot assignment came out 64/36 rather than balanced and the raters show a small slot preference, so we also report a slot-stratified estimate, which moves the pooled figure to +0.03 and each rater’s by at most 0.03. We use the unadjusted numbers throughout so that every figure in Table 14 is on one estimator, and release the stratified values with the data. Specificity behaves the same way, pooled -0.00 ($p = 0.87$), and the helpful rate is flat at 94% in both arms. On the evidence of this design we cannot claim that supplying the typed specification improves the response a user receives. We should be equally careful in the other direction. With 100 pairs and the observed spread, the smallest effect this design could detect at 80% power is about a fifth of a Likert point, so what we have ruled out is a large effect, not a small one. Detecting a gain the size of the one we measured would need on the order of 2,500 pairs. The honest statement is that the effect, if any, is small enough that this study cannot see it, not that it is zero.

The raters disagree with each other, and that is the interesting part. Table 14 breaks the result out. The lead author, who designed the system, prefers the IssueSpec arm decisively: +0.53 quality ($p = 2 \times 10^{-11}$, 59 wins, 5 losses, 36 ties) and +0.39 specificity ($p < 10^{-7}$). The two independent raters lean the other way, -0.26 and -0.18 , neither significant at $\alpha = 0.05$ ($p = 0.06$ and 0.09). Inter-rater reliability is correspondingly low, quadratic-weighted κ of 0.03, 0.04 and 0.50, against 0.97 in the discarded template-based round. High agreement there was a symptom of the confound: when one arm is visibly broken, raters agree easily. Here they do not.

Table 14: Stage-4 re-run, same model with and without the IssueSpec, 100 paired reviews, three raters, all 200 rows scored by each. All figures are unadjusted, so the gain column equals the difference of the two condition columns; slot-stratified values are in the released JSON and differ by at most 0.03. The pooled effect is null, and the disagreement between the lead author and the two independent raters is larger than the effect being measured.

rater	no_spec	full	gain	95% CI	Wilcoxon p
lead author	4.04	4.60	+0.56	[+0.44, +0.68]	2×10^{-11}
independent rater 1	3.87	3.58	-0.29	[-0.56, -0.01]	0.06
independent rater 2	3.57	3.42	-0.15	[-0.33, +0.03]	0.09
pooled	3.83	3.87	+0.04	[-0.10, +0.18]	0.38

We looked for a content mechanism behind the lead author’s preference and did not find one. Correlating each rater’s per-review preference against the spec-derived features that the IssueSpec should introduce, named devices, reproduction steps, and component mentions, produces no correlation that survives inspection for any rater; the largest is $\rho = 0.19$ ($p = 0.06$) between the lead author’s preference and reply length. So we can describe the disagreement but not explain it. Two readings remain open. The lead author may be perceiving triage value that a general reader has no reason to weigh, in which case the right evaluation asks developers rather than readers. Or the preference may be expectancy on the part of the person who built the system. This design cannot separate those, and we are not going to assert the flattering one.

Why we believe the raters this time, and what the check does and does not show. In the discarded round the two additional raters deviated from the lead author on only 16 and 19 of 400 rows, those deviation sets did not overlap on a single row, and every deviation was exactly one point. We initially read the zero

overlap as the damning fact. It is not: with deviations that rare, the hypergeometric probability of zero overlap under perfect independence is 0.45, so zero overlap is the single most likely outcome and carries almost no information. The discriminating observation is the second one. Every deviation, in both sheets, had the same magnitude, which is what a vector perturbed by a fixed step produces and not what human disagreement produces. In the reported round the raters differ from the lead author on 133 and 145 of 200 rows, the deviation sets overlap on 101 where chance predicts about 96, and the deviations take four distinct magnitudes from one to four points. We release the check with the data, and we state its weak prong plainly, because a check offered as a contribution should come with its own limits attached.

What survives. Less than we first thought. The claim that the typed IR carries downstream response quality does not survive, and neither does the retrieval finding we had treated as the surviving half: it was measured against the prompt baseline rather than a no-context arm, and against the no-context arm it is -0.07 at $p = 0.90$. Both sat in the discarded round. Stage 4 leaves us with a null and nothing else. Stage 3’s contribution, producing tracker-ready structured records at corpus scale, stands on its own evidence in Section 5.1; it no longer has a demonstrated downstream payoff in response generation.

5.3 Aligning the IR-Aware Generator at Scale

Five policies were trained on distilGPT2 (82M parameters, 400 SFT samples, 296 KTO pairs, 100 DPO pairs): SFT-base, KTO [20], DPO [50], a constrained-proxy variant, and Lagrangian Constrained PPO [14, 55]. Head-to-head on a 100-review test set, per-policy BLEU-1 / ROUGE-L / BERTScore F1 are: SFT-base 0.090 / 0.097 / 0.802; KTO 0.068 / 0.074 / 0.792; DPO 0.084 / 0.088 / 0.800; **constrained-proxy 0.137 / 0.131 / 0.806**; Lagrangian PPO 0.087 / 0.091 / 0.798. The constrained proxy leads three of four metrics (BLEU-1, ROUGE-L, BERTScore F1; the rule-based safety-violation rate is uninformative here because compliance is satisfied at initialization), with +52% BLEU-1 over SFT-base. KTO underperforms here, consistent with its small-data assumption breaking on a 296-pair training set.

The dual-objective claim is not demonstrated, and we report that as the result. At distilGPT2 scale the compliance scorer is satisfied at initialization ($C = 0.94 \geq \tau$), so the Lagrange multiplier λ collapses to zero and the update reduces to single-objective optimization. Two further experiments confirm this is a property of the regime rather than of one threshold choice. A re-run that replaces the permissive scorer with the four operational compliance checks and tightens τ from 0.50 to 0.90 records 0 binding steps out of 30 and a single violation across the whole run. A sweep over $\tau \in \{0.0, 0.5, 0.7, 0.9, 1.0\}$ leaves the pass rate at 100% for four of the five policies at every threshold, with the first binding τ undefined for those four; only SFT-base ever dips, to 99% at $\tau \geq 0.9$. A model of this size cannot plausibly violate the rubric, so the constraint has no work to do and the CMDP machinery is untested under binding conditions. The +52% BLEU-1 figure for the constrained proxy is therefore a single-objective result, not evidence of a quality-versus-compliance trade-off, and we do not claim it as one.

The constraint binds once the policy can violate it. The obstacle is not parameter count, it is capability: distilGPT2 never produces the fluent developer-relations phrasing the rubric penalises, so it cannot violate and the constraint has nothing to hold back. We measured this directly. Under the same operational scorer, 5.2% of the paper’s own Stage-4 generations trip a check, every one of them by over-promising, and a zero-shot probe of Qwen2.5-1.5B-Instruct in the same reply setting gives an indistinguishable 5.0%. distilGPT2 gives roughly zero. We therefore re-ran the Lagrangian loop with Qwen2.5-1.5B-Instruct as the policy, LoRA adapters over the attention projections, the frozen base as the KL reference, and a zero-tolerance threshold $\tau = 1.0$; the scorers, the dual-ascent rule, and the batch size are unchanged from the distilGPT2 run. The behaviour inverts. Over 90 steps the constraint binds on 13 of them (14.4%), and λ leaves zero, rises monotonically to 0.65, and never collapses back. A 30-step run under the same setup reaches $\lambda = 0.125$, so the multiplier scales with the horizon rather than saturating. Split by thirds, the trajectory shows the dual ascent responding to violations as they arrive: 1 binding step and $\lambda = 0.05$ in the first thirty, 7 and $\lambda = 0.40$ in the second, 5 and $\lambda = 0.65$ in the third. The qualitative claim that the CMDP degenerates to single-objective optimization is therefore a property of the proof-of-concept policy, not of the formulation.

What is and is not shown. The two objectives visibly pull against each other. Across the run quality rises from 0.565 to 0.590 while compliance slips from 0.997 to 0.989, and λ climbs in response, which is the dual mechanism doing exactly what it is meant to do when the policy drifts toward the reward at the constraint’s expense. What we cannot yet show is the other half of the story, the multiplier growing large enough to pull the policy back into compliance. At a 5% base violation rate and a batch of four, roughly one violating sample arrives every seven steps, and 90 steps of LoRA updates at this learning rate do not accumulate enough constraint pressure to reverse the drift. So the correct statement is narrower than a confirmed RQ3 and wider than the earlier null: the dual objective engages, the multiplier responds monotonically to violations, and the tension between the objectives is measurable, while closing the loop back to compliance needs a longer horizon or a higher base violation rate than a 1.5B instruct model supplies.

What this layer contributes is the CMDP formulation instantiated over four domain-specific dev-rel constraints, an end-to-end trainer that runs, the threshold sweep that establishes the non-binding regime, the capability probe that explains it, the binding-constraint re-run, and the released testbed.

5.4 Cluster Quality and Purity

Stage 2 was redesigned from a flat UMAP+HDBSCAN baseline (194 clusters, mean size 51.5) to the three-layer KG-hierarchical design (605 sub-clusters, mean size 3.0). We compute two complementary validity axes. *Intrinsic geometry*, Davies-Bouldin [15], Calinski-Harabasz [8], silhouette [53], follows standard IR/clustering practice [56, 58, 59]. *Extrinsic cluster purity* [2, 56] uses a Y/P/N weighted-purity rubric, $\text{purity}_w = (|Y| + 0.5|P|) / (|Y| + |P| + |N|)$, with Y = all 5 share theme, P = 3–4 of 5 share, N = fewer than 3

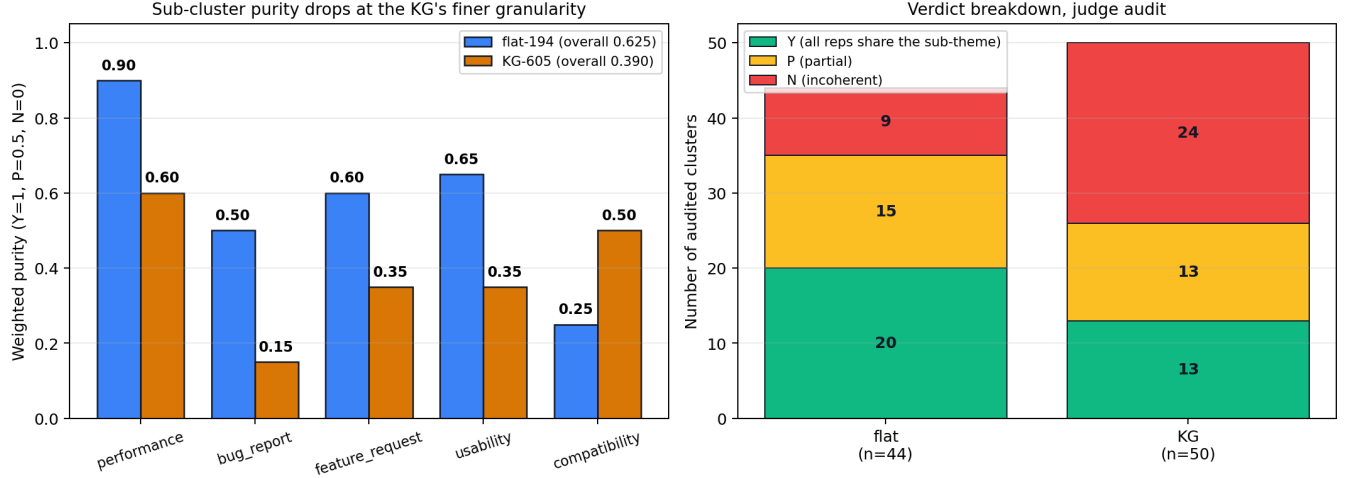


Figure 3: Extrinsic cluster purity for the flat baseline and the KG hierarchy, from the saved judge audit. Left: weighted purity per issue type, flat-194 against KG-605. The KG loses on four of five types, which is the finer-granularity cost we report rather than a result in its favour. Right: the underlying Y/P/N verdict counts behind the two overall numbers, 0.625 for flat and 0.390 for the KG. This figure is the evidence for the A2 ablation and for the claim in Section 6.1 that the KG’s contribution is prioritization, not cluster quality.

Table 15: Stage 2 designs head-to-head, comparing Pure HDBSCAN (without UMAP), original flat UMAP+HDBSCAN, and KG-hierarchical. Intrinsic metrics (Davies-Bouldin [15], Calinski-Harabasz [8], silhouette [53]) follow standard IR/clustering convention [56, 58, 59]. The Y/P/N weighted-purity rubric is a discrete-coding instantiation of the extrinsic cluster-purity metric [2, 56]. The count-controlled flat baseline (agglomerative-605) is reported separately in the A1b ablation (§5.5).

metric	Pure HDB	Flat U+H	KG hier.
n_clusters	3 (deg.)	194	605
mean cluster size	~16,933	51.5	3.0
Davies-Bouldin (↓)	n/a	12.15	2.24
Calinski-Harabasz (↑)	n/a	0.98	1.85
Y/P/N purity (lead-author)	n/a	0.66	queued
Y/P/N purity (Qwen-judge)	n/a	0.625	0.390

share, following inter-rater conventions compatible with Landis-Koch [39].

KG-hierarchical achieves 5.4× lower Davies-Bouldin and 1.9× higher Calinski-Harabasz than the flat baseline (Table 15), at the cost of finer per-cluster Y/P/N purity (0.39 vs. 0.625 Qwen-judge). By Landis-Koch [39] convention adapted to weighted purity, the flat baseline’s 0.66 lead-author purity sits in the substantial-agreement band (≥ 0.61) and KG-605’s 0.39 in the fair band (≥ 0.21), reflecting the granularity-vs-coherence trade-off the A1b ablation makes explicit.

Count-controlled diff from original. A count-controlled ablation (A1b) disambiguates two evaluation axes that prior cluster-quality work tends to conflate: intrinsic geometry and practitioner-side prioritization (centrality-weighted coverage). On 1,815 representative reviews (3 per sub-cluster across 605 sub-clusters), an agglomerative clustering forced to 605 clusters reaches DB = 1.12, CH = 4.21, silhouette = +0.148, against KG-605’s 2.24 / 1.85 / −0.234. The 5.4× DB advantage in the headline against flat-194 is therefore

primarily a cluster-count effect, not a KG-structure effect: on geometry, count-matched agglomerative clustering ties or beats the KG.

Where the KG actually wins. On the prioritization axis the story reverses. Of the 10,534 aspect nodes in the KG, the top five by PageRank account for 74% of all review-aspect mentions, and the top ten account for 96%. A developer triaging in PageRank order completes three-quarters of the complaint volume after reading five aspect buckets, whereas random aspect ordering would require thousands of buckets to reach the same coverage, a roughly $10^3 \times$ concentration ratio. The KG additionally enables per-aspect drill-down, distinguishing battery→Samsung-drain from battery→fast-charge.

5.5 Ablations

An ablation removes one component and re-measures, so that a component’s contribution is demonstrated rather than assumed. Seven were designed and six were run; the seventh is deferred for a reason given below. Each one below states what it removes, what the removal is supposed to test, what happened, and what we conclude, because three of the seven came back against the design and a summary line would hide that. Table 16 collects the verdicts.

A1, remove the knowledge graph and feed flat clusters into Stage 3. This asks whether Stage 3 reads the graph or merely the group. Stage 3 quality does not change, because the translator consumes a cluster centroid and its representative reviews, not the graph edges. The KG is therefore not a Stage 3 dependency, which narrows where its value could lie.

A1b, compare the KG against flat clustering matched to the same number of clusters. This is the reviewer-prompted version of A1 and it exists because of a confound. The KG produces 605 sub-clusters and the flat baseline produces 194, and intrinsic metrics move with cluster count on their own, so any comparison between the two conflates structure with granularity. Re-running

Table 16: Ablation summary. “Removing this hurts” is the only outcome that justifies a component. Three of the seven do not clear that bar, and the paper treats them as findings rather than as inconveniences.

id	what is removed	effect of removing it	verdict
A1	the KG, feeding flat clusters to Stage 3	Stage 3 quality unchanged	no effect
A1b	the KG, against flat clustering matched to the same 605 clusters	flat wins on all three intrinsic metrics	against the KG
A2	hierarchical sub-clustering	purity 0.625 flat vs 0.390 KG	against the KG
A3	taxonomy grounding, so the LLM writes free-form	strict fill 0.96 $\rightarrow$ 0.69; per-type structure to zero	justifies removal
A4	HITL-2 review of specs	no change to benchmark scores	no effect at benchmark time
A5	retrieval, keeping the IssueSpec	all automatic metrics move $< 1\%$	against RWF, single-stream RLHF feedback
A6	the IssueSpec, keeping retrieval	quality 3.87 $\rightarrow$ 3.83 (-0.04 , $p = 0.38$)	no measurable effect (§5.2)
A7	single-stream RLHF feedback	not run	deferred

agglomerative clustering forced to exactly 605 clusters removes the confound. The flat version then wins on all three metrics, Davies-Bouldin 1.12 against 2.24 (lower is better), Calinski-Harabasz 4.21 against 1.85 (higher is better), and silhouette +0.148 against -0.234 (higher is better, and a negative value means the average point sits closer to a neighbouring cluster than to its own). The honest conclusion is that the KG does not produce geometrically better clusters. What it produces is an ordering, through PageRank centrality over the review-aspect graph, and prioritization is the only cluster-layer claim we make.

A2, remove hierarchical sub-clustering. Weighted Y/P/N purity, judged on five representative reviews per cluster, is 0.625 for the flat 194 and 0.390 for the KG’s 605. Finer granularity costs coherence here rather than buying it. Taken with A1b this is the second result against the cluster layer, and we report both.

A3, remove taxonomy grounding. The LLM writes a free-form issue description instead of filling a routed template. Strict template-fill falls from 0.96 to 0.69, and the per-type structural columns fall to exactly zero, because an unrouted generator never emits a reproduction-step list or an As-a/I-want/So-that triple at all when nothing asks it to. This is the ablation that isolates the paper’s central mechanism. The gain is not the language model, it is the routing.

A4, remove the HITL-2 spec review. Benchmark scores do not move, and the reason is procedural rather than substantive. HITL-2 is a production-time gate; the 100 specs used in the evaluation were scored before any human correction, so the headline numbers already describe pre-HITL output. The ablation therefore says nothing about whether HITL-2 helps in deployment, and we do not claim that it does.

A5, remove retrieval and keep the IssueSpec. BLEU-1 moves by -0.006 , ROUGE-L by -0.008 , and BERTScore F1 by -0.004 , all under one percent absolute. Two caveats belong with this result. First, A5 was scored on automatic metrics only, while A6 was scored by humans, so the pair is not measured on one instrument and the contrast between them is directional rather than a like-for-like comparison. A human-rated no-retrieval condition is the missing cell. Second, we withdrew the companion claim that retrieval without the IR scores below a no-context baseline. That comparison used the prompt baseline, which is not a no-context condition, and against the arm that is, the difference

is -0.07 at $p = 0.90$. So A5’s reading is simply that retrieval makes no measurable difference here, with no accompanying story about it doing harm.

A6, remove the IssueSpec and keep retrieval. This ablation is superseded. Under the confounded design it showed quality falling from 4.59 to 2.24; under the corrected same-model design (Section 5.2) removing the IssueSpec changes quality by -0.03 , which is not distinguishable from zero. A5 and A6 together now say something different from what we first concluded: neither retrieval nor the typed intermediate produces a measurable change in the reply a user receives, and the Stage 3 to Stage 4 coupling we claimed is not supported.

RWF, single-stream RLHF feedback. Not run. It was designed to test whether quality and compliance feedback can share one channel, which only becomes a meaningful question once the constraint binds and the two signals actually compete. Section 5.3 reaches the first half of that condition and not the second, so A7 stays deferred rather than reported.

6 Discussion

6.1 What Carries the Result, and What Does Not

A five-stage pipeline invites the question of whether every stage earns its place. Our own ablations say it does not, and separating the two groups is more useful to a reader than defending the whole.

Load-bearing. Two components carry the measured gains. Type-routed templating is the principal driver of Stage 3 quality: removing taxonomy grounding (A3) drops strict template-fill from 0.96 to 0.69, and the structural per-field metrics collapse to zero because an unrouted generator does not emit the reproduction-step list or the As-a/I-want/So-that triple at all. By contrast the typed IR is *not* a demonstrated driver of Stage 4 quality: under a same-model comparison removing it moves quality by 0.04 points ($p = 0.38$, Section 5.2), and removing retrieval instead (A5) moves every automatic metric by less than one point absolute. Our earlier reading of the retrieval arm does not survive either, and the error was ours to make twice. We first reported that retrieval alone was indistinguishable from a no-context baseline, then corrected that to significantly worse. Both statements compared the retrieval arm against the *prompt* baseline, which is not a no-context condition; it carries a developer-relations instruction. Against the actual no-context arm, which sees the review text only, retrieval scores 2.24 against 2.31, a difference of -0.07 at $p = 0.90$. And all three arms come from the discarded template-composer round, so none of them supports a claim in any direction. Nothing about retrieval survives from Stage 4 beyond the null.

Not load-bearing, and reported as such. The knowledge graph does not win on intrinsic cluster geometry. Under count matching (A1b), flat agglomerative clustering beats the KG on all three intrinsic metrics, and sub-cluster purity is lower at the KG’s finer granularity (A2). The KG’s defensible contribution is centrality-based prioritization, that is, which issues a developer should see first, not tighter clusters; we make no cluster-quality claim for it. We should be straight about how far even that goes. PageRank runs on the bipartite review-aspect graph, and nothing in our results shows it

needs the hierarchical sub-clustering layer to do so. A flat clustering with the same aspect graph would plausibly support the same ranking, and we did not test that configuration. Stage 2 as built is therefore justified in its prioritization role and unjustified in its sub-clustering role, and a reader designing a lighter system should read it that way. Stage 5 sits in between. As Section 5.3 reports, the compliance constraint does bind once the policy is capable of violating it, and the dual update engages rather than collapsing, but no quality-versus-compliance trade-off is demonstrated at the scale we can train at. It is released as a testbed with a partial result. SPECov falls in the same group and for a sharper reason: tested against human faithfulness judgements it shows no correlation at all (Section 5.1), so we withdraw it as a measure rather than defend it.

What this means for the framework claim. The contribution that survives ablation is the typed IR and the type-routed templating that produces it, together with the verified-anchor procedure that makes the IR producible from noisy labels at corpus scale. Stages 2 and 5 are supporting infrastructure with honestly mixed evidence. A reader who wants the smallest system that reproduces the headline result needs Stage 1, Stage 3, and a generator conditioned on the resulting specification. That is less than Stage 4 as we built it: A5 shows the retrieval layer inside Stage 4 contributes under one percent on automatic metrics, so the part of Stage 4 that earns its place is the conditioning, not the retrieval.

6.2 Implications

Three claims travel beyond app reviews. *First*, LLM annotation noise on software-engineering text is structurally biased rather than independent-and-identically-distributed: the 25% mislabeling rate is concentrated on minority classes the LLM is blind to, not spread uniformly across the corpus. This is a finding about LLM-as-annotator failure modes that bears on any application using LLM labels for structured-text annotation. *Second*, a caution rather than a claim. We expected a typed structural intermediate to be what does the load-bearing work in spec-aware response generation, and reported it as such. Section 5.2 withdraws that. The only downstream movement we can still point to is +0.12 on a rule-based scorer from agentic critique-revise iteration, on ten reviews, which is too small a sample to build on. What generalises is the negative lesson: a structured intermediate can be demonstrably better as an artifact and still leave the generated text indistinguishable, and pipelines of this shape should measure the two separately. *Third*, the count-controlled cluster-validation ablation (A1b) delivers a self-contained methodological contribution: whenever a new clustering design changes cluster count and structure simultaneously, intrinsic geometry metrics (DB, CH, silhouette) conflate the two effects. The minimum reporting standard our analysis suggests is to (a) compare against a count-matched baseline on the same intrinsic metrics, and (b) pair that with a domain-utility metric (PageRank coverage in our case). This pattern applies directly to clustering-design comparisons in IR, software engineering, and NLP.

6.3 Threats to Validity

Construct. Template-fill measures coverage rather than content correctness, and Section 5.1 shows that SPECov does not fill that gap: it is a lexical-grounding proxy that does not track human faithfulness judgement [26, 31, 37, 42]. The paper therefore has no validated content-correctness measure for Stage 3, and the five-dimension rubric carries that burden alone. NLI- and QA-based faithfulness measurement [19, 29, 38] is the clear next step. *Internal.* The count-versus-structure confound is exposed and resolved by the A1b ablation, which reframes C3’s contribution as centrality-based prioritization rather than intrinsic geometry. A second internal threat we did not remove is that the 100 clusters carrying the Stage 3 and Stage 4 results were produced by the KG pipeline, the same layer A1b and A2 find wanting. Since flat clustering scores higher on purity (Figure 3), the headline gains may be measured on a slightly worse partition than a simpler system would produce, which would understate rather than inflate them. Re-running Stages 3 and 4 over flat clusters would settle it and is the cheapest open experiment in the paper. *External.* RRGGen contains 58 anonymized Android apps; V5 transfers zero-shot to the Maalej label space. Stage 2–4 cross-corpus transfer remains to be validated. The headline LLM is Claude Opus, with Llama-3.3-70B and Qwen2.5-{3B, 1.5B} as cross-family bias controls; GPT-4o and Gemini-1.5-Pro replication is queued (Section 6.5). *Conclusion.* The three-rater agreement on the classification gold ($\kappa = 0.968$ on the Y/N verdict) sits well above what free-labeling studies of app reviews report, which invites the question of whether the raters were genuinely independent. Three properties of the design account for the gap. The raters verify a supplied label rather than choose one from seven, which collapses most of the decision space; a 20-review calibration round with adjudicated disagreements precedes the main task; and the written protocol fixes eight boundary rules that carry the cases annotators most often split on. Residual disagreement is rater-specific rather than random, one rater accounts for most of the usability reassignments, which is the pattern independent annotation produces. Stage 4 sits differently. All three raters scored the complete 200-row re-run, and the gain does *not* reproduce per rater: the lead author measures +0.53 and the two independent raters −0.26 and −0.18. That disagreement, not the pooled null, is the honest summary of what we know about Stage 4. *Reliability, and which α belongs to what.* Three agreement statistics appear in this paper and they measure different things, so a single “moderate α ” reading would be wrong. The classification gold is annotated by three humans and reaches Fleiss $\kappa = 0.968$ on the verdict and 0.979 on the seven-class label. The Stage-4 response set that we report is rated by the same three humans on all 200 rows, with quality weighted κ of 0.03, 0.04 and 0.50. That is low, and it is the number to quote: the 0.970 to 0.986 range belongs to the discarded round, where one arm was visibly broken and agreement was easy. The one moderate figure, $\alpha = 0.451$, comes from a different and deliberately weaker panel: a cluster-purity check whose three raters are the lead author plus two Qwen2.5-3B prompts, following the LLM-as-rater convention of Gilardi et al. [25]. It bounds how far an LLM rater tracks a human on a subjective sub-theme judgement; it is not the reliability of any headline result, and no claim in this paper rests on it alone. *Statistical.* Experiment 2 now

reports a null, so the relevant question is power rather than correction for multiplicity. With 100 paired reviews and the observed per-review spread, the 95% bootstrap interval $[-0.10, +0.18]$ excludes effects larger than about a fifth of a Likert point in either direction. We can therefore rule out a large effect, not a small one; an effect of a tenth of a point would need several times this sample.

6.4 Reproducibility and Extensibility

Every numerical claim in Section 5 is backed by a saved data file and runs end-to-end from raw RRGGen to Stage 4 output in approximately six hours on a single GPU-equipped workstation. The release bundles:

- V1–V5 classifier checkpoints + cleanlab correction procedure;
- the 5,230-review verified anchor and 490-review classification gold;
- both Stage-4 rating rounds, the discarded 400-row one and the reported 200-row re-run, together with the same-model generated replies and the rating-provenance check that distinguishes them;
- the count-controlled A1b output;
- the SPECov scorer plus the 60-spec, three-rater human faithfulness ratings and the validation analysis;
- the CMDP-RLHF testbed (KTO, DPO, constrained-proxy, and Lagrangian Constrained PPO trainers);
- the 100-cluster benchmark, with the 99-review LLM-assisted purity subsample ($\alpha = 0.451$) kept separate from the all-human agreement sets;
- a pinned-environment Docker image (single-command build on any CUDA 12.x host);
- a nineteen-segment verifier that recomputes the numerical claims in Section 5 from the saved data files, including the corrected Stage-4 re-run, the three-rater agreement statistics, the cross-family rubric, the SPECov validation, the binding-constraint run, and the generation-variance sweep. Two tables are not covered by it, the per-class F1 breakdown and the multi-model template-fill row, and we say so rather than claim complete coverage. The discarded Stage-4 round retains its own segment, labelled as withdrawn.

The pipeline is composable by construction: because each stage emits a typed intermediate, components (classifier, clusterer, taxonomy template, alignment trainer) can be substituted independently without breaking downstream contracts. Future researchers can adopt or replace any single component while keeping the rest of the IR-centered evaluation harness intact. Deterministic components (clustering, classifier training) use fixed seeds; the LLM-dependent stages (annotation, IR generation, and response generation) reproduce statistically rather than bit-identically. We measured how large that sampling variance is. Regenerating Stage 3 specs three times over the same 15 stratified clusters with the local Qwen2.5-3B generator at temperature 0.7, changing only the seed, gives strict template-fill 0.685, 0.700, and 0.634 (mean 0.673, sd 0.029, range 0.067) and SPECov 2.73, 2.67, and 3.00 (mean 2.80, sd 0.144, range 0.333), with no JSON parse failures in any run. Run-to-run noise on template-fill is therefore around three points on a 0–1 scale for this generator and prompt. That is the yardstick a

reader should hold up against the differences we report. The gaps the paper rests on clear it comfortably: 0.96 versus 0.69 for taxonomy grounding, and 0.96 versus 0.48 across the model families. The withdrawn Stage-4 effect illustrates the opposite case, and is worth stating in these terms: at +0.04 it sits an order of magnitude below this noise floor, which is another way of saying there is nothing there to measure. Differences of a few hundredths between adjacent conditions, such as the 4.09 versus 3.96 at the top of the cross-family rubric, do not clear it and should not be read as a stable ordering on their own. Two limits remain: the headline Claude generations were not repeated, because that needs API budget rather than new code, and Stage 4 response generation was not repeated either. The released script runs both sweeps unchanged.

6.5 Future Work

Five extensions map directly to the limitations above; each starts from a resource released with this paper.

(0) Replace SPECov and widen the cross-family comparison. The SPECov validation in Section 5.1 came back negative, so Stage 3 needs a faithfulness measure that actually works. The natural replacement is entailment-based: score each spec field against the source reviews with an NLI model [29, 38] or a question-generation-and-answering pipeline [19], then validate that measure against the same 60-spec human ratings we release here, which now serve as a ready-made benchmark. Second, the cross-family rubric comparison (Table 8) rests on 14 matched bug-report clusters for the four-way panel, because that is the extent of the overlap between the four generators’ spec sets, and it is scored by a judge from one of the four families. Generating the missing spec types for the two Qwen models and re-scoring with a judge outside all four families would remove both limits. Both are runs rather than redesigns.

(1) Ask the right raters, then scale. The Stage 4 re-run carries three annotators on all 200 rows and returns a null with the raters disagreeing in direction. Before growing the sample, the design should change: the reply from a triage pipeline is consumed by a developer deciding what to do, not by a general reader, so the next round should recruit raters with software-engineering experience and no stake in the system. Scaling from 100 clusters to 200–300, with three annotators per IssueSpec and Krippendorff’s α at the IssueSpec level, follows that rather than precedes it.

(2) CMDP RLHF at a scale where the trade-off shows. The binding-constraint re-run in Section 5.3 establishes that the dual objective engages; what it does not show is a trade-off, because a 5% base violation rate over 30 steps supplies too little signal. The next run wants a longer horizon and a policy with a higher base violation rate, on an 8B-class model. *Original plan.* Where the compliance constraint is expected to bind, activating the Lagrange multiplier and making the Pareto-dominance comparison meaningful.

(3) Frontier-family replication. GPT-4o and Gemini-1.5-Pro on Stages 3–4. The current paper closes the Meta-family slice with Llama-3.3-70B and the Alibaba slice with Qwen2.5-{3B, 1.5B}.

(4) Cross-corpus transfer. Stages 2–4 on additional review corpora; V5 already transfers zero-shot to Maalej [41].

(5) **Agentic Planner extension to Stage 4.** The Planner dispatches a different sub-agent profile per issue type: defect-repair for `bug_report` via SWE-Agent [62] or RepairAgent [6], scaffolding for `feature_request`, profiler-driven optimization for performance, a Nielsen-heuristic checker for usability, and a device-OS test-matrix executor for compatibility. A spec-level stub already runs the dispatch across all five issue types; end-to-end patch evaluation on real repositories through HyperAgent [49], measured by patch acceptance and test-pass rate, is the next step.

7 Conclusion

We proposed **IssueSpec**, a knowledge-graph-grounded typed intermediate representation that closes the five existing research gaps identified in Section 1: classification stops at the label, response generation skips the structured intermediate, agentic software engineering assumes the spec exists, knowledge graphs emit no typed schema, and RLHF has not addressed dev-rel compliance. We set out to show that the typed IR is the load-bearing component of spec-aware response generation, and we cannot. An isolation experiment that holds the generator fixed moves human-rated quality by +0.04 Likert points ($p = 0.38$, three raters, 100 paired reviews), and the raters disagree in direction. The +2.35 we previously reported came from a comparison whose two arms were written by different template composers. Nor does the negative half of our original hypothesis survive. We had reported that retrieval alone falls below a no-context baseline; against the arm that actually has no context it scores 2.24 against 2.31 at $p = 0.90$, and that comparison sits inside the discarded round in any case. Stage 4 yields one usable result, which is the null. A verified-anchor procedure makes the IR producible at corpus scale from noisy LLM labels, lifting Cohen’s κ from 0.163 to 0.592, and to 0.616 on the strictly held-out gold, with 88.66% classifier endorsement of corrections. Cross-family replication on four LLMs (Claude, Llama, Qwen-3B, Qwen-1.5B) preserves the rank order, ruling out single-model bias. Two findings travel beyond app reviews: LLM annotation noise on software-engineering text is structurally biased rather than random, and that a structured intermediate can be demonstrably better as an artifact while leaving the text generated from it indistinguishable, a distinction that pipeline evaluations of this shape should measure separately. We release the IssueSpec schema, the verified anchor, the SPECov scorer together with the human ratings that show it does not measure faithfulness, the count-controlled cluster-validation ablation, the structured-RAG architecture, the CMDP-RLHF trainers, and the benchmark dataset to support follow-on research.

GenAI Usage Disclosure

We disclose all uses of generative AI in this work in line with CIKM 2026 policy. *Writing and code assistance.* Anthropic’s Claude (Opus 4.7) [3] assisted with grammatical refinement, caption polishing, LaTeX formatting, and reviewing implementation patterns and statistical-test code. All pipeline code, experimental scripts, and analysis were written by the human authors. Every structural, framing, experimental-design, and claim decision was made by

the authors, and every reported number was produced by author-written code run under human supervision and verified against saved data files. *Models inside the system.* Several generative models are themselves pipeline components and are reported as such in the body: Claude (Opus 4.7) is the headline Stage 3 generator (Table 11); Llama-3.3-70B-Instruct (via Groq) and Qwen2.5-{3B, 1.5B}-Instruct are cross-family Stage 3 comparators (Table 9); Qwen2.5-3B-Instruct is the LLM-as-judge for the five-dimension rubric and the Y/P/N cluster-purity audit, both calibrated against the lead author (Tables 10, 15). The V2 corpus labels in Section 5.1 are LLM-generated and explicitly treated as noisy, which is precisely what motivates the verified-anchor procedure of Stage 1. *Human verification.* No reported numerical result, expert annotation, or ground-truth label was automatically generated by AI without human verification. The 5,230-review verified anchor, the 490-review classification gold, the 99-review three-rater purity ratings, and the 20 lead-author IssueSpec references were all produced and reviewed by human annotators (Section 4).

References

- [1] Eitan Altman. 2021. *Constrained Markov decision processes*. Routledge.
- [2] Enrique Amigó, Julio Gonzalo, Javier Artilles, and Felisa Verdejo. 2009. A comparison of extrinsic clustering evaluation metrics based on formal constraints. *Information retrieval* 12, 4 (2009), 461–486.
- [3] Anthropic. 2025. Claude (Opus 4.7), Large Language Model. <https://www.anthropic.com/claude>. Used as a writing and formatting assistant during manuscript preparation..
- [4] Akari Asai, Zeqiu Wu, Yizhong Wang, Avi Sil, and Hannaneh Hajishirzi. 2024. Self-rag: Learning to retrieve, generate, and critique through self-reflection. In *International conference on learning representations*, Vol. 2024. 9112–9141.
- [5] Gagan Bansal, Besmira Nushi, Ece Kamar, Walter S Lasecki, Daniel S Weld, and Eric Horvitz. 2019. Beyond accuracy: The role of mental models in human-AI team performance. In *Proceedings of the AAAI conference on human computation and crowdsourcing*, Vol. 7. 2–11.
- [6] Islem Bouzenia, Premkumar Devanbu, and Michael Pradel. 2025. Repairagent: An autonomous, llm-based agent for program repair. In *2025 IEEE/ACM 47th International Conference on Software Engineering (ICSE)*. IEEE, 2188–2200.
- [7] Sergey Brin and Lawrence Page. 1998. The anatomy of a large-scale hypertextual web search engine. *Computer networks and ISDN systems* 30, 1-7 (1998), 107–117.
- [8] Tadeusz Caliński and Jerzy Harabasz. 1974. A dendrite method for cluster analysis. *Communications in Statistics-theory and Methods* 3, 1 (1974), 1–27.
- [9] Oscar Chaparro, Jing Lu, Fiorella Zampetti, Laura Moreno, Massimiliano Di Penta, Andrian Marcus, Gabriele Bavota, and Vincent Ng. 2017. Detecting missing information in bug descriptions. In *Proceedings of the 2017 11th joint meeting on foundations of software engineering*. 396–407.
- [10] Ning Chen, Jialiu Lin, Steven CH Hoi, Xiaokui Xiao, and Boshen Zhang. 2014. AR-miner: mining informative reviews for developers from mobile app marketplace. In *Proceedings of the 36th international conference on software engineering*. 767–778.
- [11] Norman Cliff. 1993. Dominance statistics: Ordinal analyses to answer ordinal questions. *Psychological bulletin* 114, 3 (1993), 494.
- [12] Mike Cohn. 2004. *User stories applied: For agile software development*. Addison-Wesley Professional.
- [13] Jacek Dąbrowski, Emmanuel Letier, Anna Perini, and Angelo Susi. 2022. Analysing app reviews for software engineering: a systematic literature review. *Empirical Software Engineering* 27, 2 (2022), 43.
- [14] Juntao Dai, Xuehai Pan, Ruiyang Sun, Jiaming Ji, Xinbo Xu, Mickel Liu, Yizhou Wang, and Yaodong Yang. 2024. Safe rlhf: Safe reinforcement learning from human feedback. In *International Conference on Learning Representations*, Vol. 2024. 50750–50777.
- [15] David L Davies and Donald W Bouldin. 1979. A cluster separation measure. *IEEE Transactions on Pattern Analysis and Machine Intelligence PAMI-1*, 2 (1979), 224–227.
- [16] AJW De Vink, Natalia Amat-Lefort, and Lifeng Han. 2025. ReviewGraph: A Knowledge Graph Embedding Based Framework for Review Rating Prediction with Sentiment Features. In *2025 IEEE International Conference on Knowledge Graph (ICKG)*. IEEE, 43–50.
- [17] Janez Demšar. 2006. Statistical comparisons of classifiers over multiple data sets. *Journal of Machine learning research* 7, Jan (2006), 1–30.

- [18] Andrea Di Sorbo, Sebastiano Panichella, Carol V Alexandru, Corrado A Visaggio, and Gerardo Canfora. 2017. Surf: Summarizer of user reviews feedback. In *2017 IEEE/ACM 39th International Conference on Software Engineering Companion (ICSE-C)*. IEEE, 55–58.
- [19] Esin Durmus, He He, and Mona Diab. 2020. FEQA: A question answering evaluation framework for faithfulness assessment in abstractive summarization. In *Proceedings of the 58th annual meeting of the association for computational linguistics*. 5055–5070.
- [20] Kavin Ethayarajh, Winnie Xu, Niklas Muennighoff, Dan Jurafsky, and Douwe Kiela. 2024. Kto: Model alignment as prospect theoretic optimization. *arXiv preprint arXiv:2402.01306* (2024).
- [21] International Organization for Standardization, Software Technical Committee ISO/IEC JTC 1, Information technology. Subcommittee SC 7, and systems engineering. 2011. *Systems and Software Engineering: Systems and Software Quality Requirements and Evaluation (SQuaRE): System and Software Quality Models*. ISO.
- [22] Cuiyun Gao, Jichuan Zeng, Xin Xia, David Lo, Michael R Lyu, and Irwin King. 2019. Automating app review response generation. In *2019 34th IEEE/ACM International Conference on Automated Software Engineering (ASE)*. IEEE, 163–175.
- [23] Cuiyun Gao, Wenjie Zhou, Xin Xia, David Lo, Qi Xie, and Michael R Lyu. 2021. Automating app review response generation based on contextual knowledge. *ACM Transactions on Software Engineering and Methodology (TOSEM)* 31, 1 (2021), 1–36.
- [24] Yonatan Geifman and Ran El-Yaniv. 2017. Selective classification for deep neural networks. *Advances in neural information processing systems* 30 (2017).
- [25] Fabrizio Gilardi, Meysam Alizadeh, and Maël Kubli. 2023. ChatGPT outperforms crowd workers for text-annotation tasks. *Proceedings of the National Academy of Sciences* 120, 30 (2023), e2305016120.
- [26] Max Grusky, Mor Naaman, and Yoav Artzi. 2018. Newsroom: A dataset of 1.3 million summaries with diverse extractive strategies. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers)*. 708–719.
- [27] Emitza Guzman and Walid Maalej. 2014. How do users like this feature? a fine grained sentiment analysis of app reviews. In *2014 IEEE 22nd international requirements engineering conference (RE)*. Ieee, 153–162.
- [28] Marti A Hearst. 1999. Untangling text data mining. In *Proceedings of the 37th Annual meeting of the Association for Computational Linguistics*. 3–10.
- [29] Or Honovich, Roei Aharoni, Jonathan Herzog, Hagai Taitelbaum, Doron Kukliansky, Vered Cohen, Thomas Scialom, Idan Szepktor, Avinatan Hassidim, and Yossi Matias. 2022. TRUE: Re-evaluating factual consistency evaluation. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. 3905–3920.
- [30] Mingqing Hu and Bing Liu. 2004. Mining and summarizing customer reviews. In *Proceedings of the tenth ACM SIGKDD international conference on Knowledge discovery and data mining*. 168–177.
- [31] Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung. 2023. Survey of hallucination in natural language generation. *ACM computing surveys* 55, 12 (2023), 1–38.
- [32] Carlos E Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik Narasimhan. 2023. SWE-bench: Can language models resolve real-world github issues? *arXiv preprint arXiv:2310.06770* (2023).
- [33] Mona Erfani Joorabchi, Ali Mesbah, and Philippe Kruchten. 2013. Real challenges in mobile app development. In *2013 ACM/IEEE International Symposium on Empirical Software Engineering and Measurement*. IEEE, 15–24.
- [34] Kamaljit Kaur and Parminder Kaur. 2023. BERT-RCNN: An automatic classification of app reviews using transfer learning based RCNN deep model. *Research Square preprint* (2023).
- [35] Swetha Keertipati, Bastin Tony Roy Savarimuthu, and Sherlock A Licorish. 2016. Approaches for prioritizing feature improvements extracted from app reviews. In *Proceedings of the 20th international conference on evaluation and assessment in software engineering*. 1–6.
- [36] Omar Khattab, Keshav Santhanam, Xiang Lisa Li, David Hall, Percy Liang, Christopher Potts, and Matei Zaharia. 2022. Demonstrate-search-predict: Composing retrieval and language models for knowledge-intensive nlp. *arXiv preprint arXiv:2212.14024* (2022).
- [37] Wojciech Kryściński, Bryan McCann, Caiming Xiong, and Richard Socher. 2020. Evaluating the factual consistency of abstractive text summarization. In *Proceedings of the 2020 conference on empirical methods in natural language processing (EMNLP)*. 9332–9346.
- [38] Philippe Laban, Tobias Schnabel, Paul N Bennett, and Marti A Hearst. 2022. SummaC: Re-visiting NLI-based models for inconsistency detection in summarization. *Transactions of the Association for Computational Linguistics* 10 (2022), 163–177.
- [39] J Richard Landis and Gary G Koch. 1977. The measurement of observer agreement for categorical data. *biometrics* (1977), 159–174.
- [40] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, et al. 2020. Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in neural information processing systems* 33 (2020), 9459–9474.
- [41] Walid Maalej and Hadeer Nabil. 2015. Bug report, feature request, or simply praise? on automatically classifying app reviews. In *2015 IEEE 23rd international requirements engineering conference (RE)*. IEEE, 116–125.
- [42] Joshua Maynez, Shashi Narayan, Bernd Bohnet, and Ryan McDonald. 2020. On faithfulness and factuality in abstractive summarization. In *Proceedings of the 58th annual meeting of the association for computational linguistics*. 1906–1919.
- [43] Leland McInnes, John Healy, Steve Astels, et al. 2017. hdbscan: Hierarchical density based clustering. *J. Open Source Softw.* 2, 11 (2017), 205.
- [44] Leland McInnes, John Healy, and James Melville. 2018. Umap: Uniform manifold approximation and projection for dimension reduction. *arXiv preprint arXiv:1802.03426* (2018).
- [45] Jakob Nielsen. 1994. *Usability engineering*. Morgan Kaufmann.
- [46] Curtis Northcutt, Lu Jiang, and Isaac Chuang. 2021. Confident learning: Estimating uncertainty in dataset labels. *Journal of Artificial Intelligence Research* 70 (2021), 1373–1411.
- [47] Lawrence Page, Sergey Brin, Rajeev Motwani, and Terry Winograd. 1999. *The PageRank citation ranking: Bringing order to the web*. Technical Report 1999-66. Stanford InfoLab.
- [48] Nicholas Pangakis, Samuel Wolken, and Neil Fasching. 2023. Automated annotation with generative ai requires validation. *arXiv preprint arXiv:2306.00176* (2023).
- [49] Huy Nhat Phan, Tien N Nguyen, Phong X Nguyen, and Nghi DQ Bui. 2024. Hyperagent: Generalist software engineering agents to solve coding tasks at scale. *arXiv preprint arXiv:2409.16299* (2024).
- [50] Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D Manning, Stefano Ermon, and Chelsea Finn. 2023. Direct preference optimization: Your language model is secretly a reward model. *Advances in neural information processing systems* 36 (2023), 53728–53741.
- [51] Nils Reimers and Iryna Gurevych. 2019. Sentence-bert: Sentence embeddings using siamese bert-networks. In *Proceedings of the 2019 conference on empirical methods in natural language processing and the 9th international joint conference on natural language processing (EMNLP-IJCNLP)*. 3982–3992.
- [52] Jeanine Romano, Jeffrey D Kromrey, Jesse Coraggio, and Jeff Skowronek. 2006. Appropriate statistics for ordinal level data: Should we really be using t-test and Cohen’s d for evaluating group differences on the NSSE and other surveys. In *Annual Meeting of the Florida Association of Institutional Research*. 1–33.
- [53] Peter J Rousseeuw. 1987. Silhouettes: a graphical aid to the interpretation and validation of cluster analysis. *Journal of computational and applied mathematics* 20 (1987), 53–65.
- [54] Sunita Sarawagi. 2008. Information extraction. *Foundations and trends in databases* 1, 3 (2008), 261–377.
- [55] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. 2017. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347* (2017).
- [56] Hinrich Schütze, Christopher D Manning, and Prabhakar Raghavan. 2008. *Introduction to information retrieval*. Vol. 39. Cambridge University Press Cambridge.
- [57] Faiz Ali Shah, Kairit Sirts, and Dietmar Pfahl. 2020. Mining User Opinions to Support Requirement Engineering: An Empirical Study. In *Advanced Information Systems Engineering (CAiSE 2020) (Lecture Notes in Computer Science, Vol. 12127)*. Springer, 401–416.
- [58] Michael Steinbach, George Karypis, and Vipin Kumar. 2000. A comparison of document clustering techniques. In *KDD Workshop on Text Mining*. 525–526.
- [59] Alexander Strehl and Joydeep Ghosh. 2002. Cluster ensembles—a knowledge reuse framework for combining multiple partitions. *Journal of machine learning research* 3, Dec (2002), 583–617.
- [60] Lorenzo Villarroel, Gabriele Bavota, Barbara Russo, Rocco Oliveto, and Massimiliano Di Penta. 2016. Release planning of mobile apps based on user reviews. In *proceedings of the 38th International Conference on Software Engineering*. 14–24.
- [61] Matt Wynne, Aslak Hellesoy, and Steve Tooke. 2017. *The Cucumber Book: Behaviour-Driven Development for Testers and Developers* (2 ed.). Pragmatic Bookshelf.
- [62] John Yang, Carlos E Jimenez, Alexander Wettig, Kilian Lieret, Shunyu Yao, Karthik Narasimhan, and Ofir Press. 2024. SWE-agent: Agent-computer interfaces enable automated software engineering. *Advances in Neural Information Processing Systems* 37 (2024), 50528–50652.
- [63] Zheng Yu, Songyu Wu, Jielin Jiang, and Dongqing Liu. 2024. A knowledge-graph based text summarization scheme for mobile edge computing. *Journal of Cloud Computing* 13, 1 (2024), 9.
- [64] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric Xing, et al. 2023. Judging llm-as-a-judge with mt-bench and chatbot arena. *Advances in neural information processing systems* 36 (2023), 46595–46623.
- [65] Thomas Zimmermann, Rahul Premraj, Nicolas Bettenburg, Sascha Just, Adrian Schroter, and Cathrin Weiss. 2010. What makes a good bug report? *IEEE Transactions on Software Engineering* 36, 5 (2010), 618–643.

A Implementation Settings

Section 3 states what each stage does and why. This appendix collects the exact settings, so the method section carries the logic and the numbers stay checkable. Stage 1 fine-tunes a RoBERTa multi-label head; confident-learning correction against the verified anchor is gated by anchor confidence ≥ 0.70 and LLM probability ≤ 0.20 , with a 3×3 sensitivity grid over $(0.65, 0.70, 0.75) \times (0.15, 0.20, 0.25)$ reported in Section 5.1. The compatibility stratum is seeded with 200 template-generated synthetic reviews (device, OS, and screen templates) plus 100 keyword-mined RRGGen reviews, all human-checked. Stage 2 embeds with Sentence-BERT, reduces with UMAP, and clusters with HDBSCAN, aspect-first, producing 605 sub-clusters of mean size 3.0 against 194 flat clusters of mean size 51.5; PageRank runs on the bipartite review-aspect graph. Stage 3 routes to one of five templates and re-prompts once with dimension-level feedback on any rubric failure. Stage 4 retrieves over a 15,100-document ChromaDB index built from five sources, with the IssueSpec injected per query rather than pre-indexed. Stage 5 trains five policies on distilGPT2 (82M parameters) from 400 SFT samples, 296 KTO pairs, and 100 DPO pairs; the Lagrangian loop runs 30 steps at batch size 4 with KL coefficient 0.05, and the compliance threshold sweep covers $\tau \in \{0.0, 0.5, 0.7, 0.9, 1.0\}$. SPECov counts substantive-token overlap with the source cluster, tokens of length ≥ 5 after stop-word filtering, with quoted-string and coverage bonuses.

B Annotator Guidelines

The lead author (PhD in software engineering with prior mobile-app review-mining experience) served as primary annotator across all five annotation tasks. The 490-review classification gold was additionally annotated by two further independent human raters on the identical item set, aligned by review id; all three first completed a shared 20-review calibration round, disagreements were adjudicated before the main task opened, and the released gold label is the majority vote over the three raters. All human annotation in this work is carried out by those same three people. For Stage 4, all three raters scored all 200 rows of the reported re-run independently and blind to condition; the discarded 400-row round had the lead author as primary rater with one additional rater on a 30-pair subset, and is described here only because the paper reports its withdrawal. Separately, two Qwen2.5-3B-Instruct prompts (concise role-based and chain-of-thought) served as the second and third raters in the 99-review cluster-purity Krippendorff α calibration, which is an LLM-assisted panel and not one of the human rounds. For the 5,230-review verified anchor, annotators classified each review into one of seven Maalej-Nabil classes (bug_report, feature_request, performance, usability, compatibility, praise, other) with multi-label assignment permitted when a review carried multiple concerns; reviews below the “fairly confident” threshold were skipped and excluded from the anchor. For the 490-review classification gold standard, a stratified sample of 70 reviews per class was drawn from the working corpus, and the lead author re-labeled each review independently and blinded to the V2 LLM prediction; the 490 gold labels serve as the held-out evaluation set for Cohen’s κ progression (Table 12) and per-class

F1 (Table 13). For the 5-dimension IssueSpec rubric (all 100 specs; an earlier stratified $n = 28$ subsample, 6 per issue type plus 2 extras), each spec was scored on a 1–5 Likert scale across five dimensions: *completeness* (are all required template fields filled?), *accuracy* (do field values correctly reflect the source cluster?), *actionability* (would a developer know what to do?), *specificity* (does the spec name concrete entities rather than generic terms?), and *clarity* (is the spec easy to read and unambiguous?), with each dimension scored independently. For the reported Stage 4 evaluation (100 reviews $\times$ 2 conditions, the same model with and without the IssueSpec, 200 rows), all three raters rated *quality* (1–5), *specificity* (1–5), and *helpful* (Y/N) while blinded to the generation method; the superseded 400-row round used four template-composed conditions and is retained in the release for comparison. In both rounds condition labels were replaced with random A/B/C/D codes and unsealed only at aggregation, with the assignment persisted as spreadsheets with JSON master keys for audit. A second human rater independently scored a 30-pair (no_spec, full) subset for inter-rater calibration. For the Y/P/N cluster-purity rubric (initial 50-cluster audit; 99-cluster three-rater subsample), annotators read 5 representative reviews per cluster and assigned Y (all 5 share the same sub-theme within the cluster’s aspect), P (3 or 4 of 5 share the sub-theme), or N (fewer than 3 share); weighted purity is $\text{purity}_w = (|Y| + 0.5|P|) / (|Y| + |P| + |N|)$, with Y contributing 1.0, P contributing 0.5, and N contributing 0. All comparative evaluations were blinded throughout the rating window and unsealed only at aggregation; reviews or specs below the annotator confidence threshold were marked as “skip” and excluded from downstream analysis.